

ALGEBRA 2

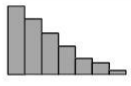
Unit II

Name:	Topic:
Main Ideas/Questions	Notes/Examples
FUNDAMENTAL COUNTING PRINCIPLE	If decision M can be made in _____ ways, then the number of ways that _____ can be made is _____.
EXAMPLES	1. The ice cream shop offers 5 different toppings. How many different ice cream cones can be made with one topping? 2. Virginia license plates consist of 3 letters followed by 4 numbers. How many different license plates can be made? 3. Nick's science test has 10 questions. How many different ways can Nick answer the test?

Name:	Topic:
Main Ideas/Questions	Notes/Examples
Theoretical Probability	- Probability is the measure of how likely an event is to occur. - The set of all possible outcomes of an experiment is called the sample space. - For equally likely outcomes, the probability $P(E)$ is the ratio of the number of favorable outcomes to the total number of outcomes.
Simple Events	1. A jar contains 32 red marbles and 28 blue marbles. What is the probability that a randomly selected marble is red?

Name:	Topic:
Main Ideas/Questions	Notes/Examples
Conditional Probability	- Given events A and B, the conditional probability of B occurring, knowing A has occurred, is denoted by $P(B A)$. - Notation: $P(B A)$ - Read as: "the probability of B given A".
Examples	1. Mitchell drew a card from a deck of playing cards. What is the probability that it is a heart, given that it is a red card? 2. A number from 1-100 is randomly selected. What is the probability that it is a perfect square, given that it is a multiple of 10?

Name:	Topic:
Main Ideas/Questions	Notes/Examples
BINOMIAL EXPERIMENTS	A binomial experiment consists of n independent trials, each with two possible outcomes: success or failure. The probability of success is p and the probability of failure is q .
EXAMPLES	1. There are only two outcomes: success or failure. 2. The probability of success is p and the probability of failure is q . 3. The trials are independent. 4. The probability of success is constant for each trial. 5. The probability of failure is constant for each trial.
Binomial Probability	If a binomial experiment has n trials, each with two possible outcomes: success or failure, and the probability of success is p and the probability of failure is q , then the probability of getting exactly x successes in n trials is given by the binomial probability formula:

Name:	Topic:
Main Ideas/Questions	Notes/Examples
Data Distribution	
Normal Distribution	- A symmetrical, bell-shaped curve. - The area under the curve is equal to 1. - The mean, median, and mode are all equal.
The Empirical Rule	In a normal distribution, approximately: 68% of the data falls within 1 standard deviation of the mean. 95% of the data falls within 2 standard deviations of the mean. 99.7% of the data falls within 3 standard deviations of the mean.

Name:	Topic:
Main Ideas/Questions	Notes/Examples
Z-SCORES	- A value that shows how many standard deviations a data value is from the mean. - When the z-score is positive, the data value is above the mean. - When the z-score is negative, the data value is below the mean.

PROBABILITY & STATISTICS

NOTES • HOMEWORK • QUIZZES • TEST

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Unit 11 - Probability & Statistics: Sample Unit Outline

	TOPIC	HOMEWORK
DAY 1	Permutations and Combinations	HW #1
DAY 2	Theoretical Probability, Independent and Dependent Events	HW #2
DAY 3	Conditional Probability	HW #3
DAY 4	Binomial Theorem, Binomial Probability	HW #4
DAY 5	Quiz 11-1	None
DAY 6	Measures of Center & Variation, Normal Distribution Curve	HW #5
DAY 7	Z-Scores, Standard Normal Distribution, Probability under the Curve	HW #6
DAY 8	Quiz 11-2	None
DAY 9	Unit 11 Review	Study for Test
DAY 10	UNIT 11 TEST	None

See sample images of the pages on the next page.

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
FUNDAMENTAL COUNTING PRINCIPLE	If decision M can be made _____ ways and decision N can be made _____ ways, then the two decisions can be made _____ ways.
EXAMPLES	- The ice cream shop offers a choice of a 3 cone sizes, 15 flavors, and 8 toppings. How many cones are possible if you can only choose one flavor and one topping? - Virginia license plates consist of three letters followed by 4 digits. How many different license plates are possible? - Nick's science test has a section with 12 true or false questions. How many different ways can Nick answer these questions? - To enter their home, the Clayton family enters a 4-digit code. How many codes are possible?
FACTORIALS 	The _____ of all natural numbers from _____ to _____. Examples: $9! =$ _____ $=$ _____ $12! =$ _____ $8! =$ _____ $\frac{17!}{11!} =$ _____ $*0! =$ _____
PERMUTATIONS 	An arrangement or line up of objects in which _____ ! Permutation Formula: ${}_nP_r =$ n = total number of objects available r = number of objects to use for the arrangement *Important Shortcut: ${}_nP_n =$
EXAMPLES	- How many ways can you arrange the letters in the word DINOSAUR? - Seven students are competing in a geography bee. How many ways can they finish in first, second, and third place?

	7. Melanie is taking four classes this semester: American History, Algebra 2, AP English, and Chemistry. How many ways can these four classes be arranged on her schedule?	
	8. There are 16 players on the baseball team. How many ways can the coach make a 9-player batting order?	
	9. There are 28 students in Mr. Miller's homeroom. How many ways can the students elect a student council representative and alternate?	
PERMUTATIONS <i>with Repetition</i>	10. How many different 10-letter arrangements are possible using the letters in the word AUTOMOBILE ?	11. How many different 9-letter arrangements are possible using the letters in the word DISAPPEAR ?
COMBINATIONS 	A group of objects in which _____ ! Combination Formula: ${}_nC_r =$ n = total number of objects available r = number of objects to use for the arrangement *Important Shortcut: ${}_nC_n =$	
EXAMPLES	12. Natalie has 16 close friends. How many ways can she choose 5 to be bridesmaids in her wedding?	
	13. There are 24 students in Kyle's kindergarten class. How many ways can he choose eight to attend his birthday party?	
	14. Abby is adopting kittens from the pet store. If there are 18 kittens, how many ways can she choose two?	
	15. There are twelve employees at the sub shop. How many ways can the manager choose four for the Sunday evening shift?	
	16. There are 85 players on the football team. How many ways can the coach choose three to represent the team in the coin toss?	

Permutation or Combination?

Directions: Recall that a **permutation is an arrangement** with a specific order, while a **combination is a group** with no specific order. Determine whether the example represents a permutation or combination, then solve.

	Example	P or C?	Answer
1	How many ways can 4 candy bars be chosen from a store that sells 30 candy bars?		
2	How many ways can 13 students line up for lunch?		
3	How many ways can you make a 3-letter arrangement out of the letters in the word TRAPEZOID .		
4	How many ways can you choose 2 books from a shelf of 40 books?		
5	How many ways can 12 swimmers finish in first, second, and third place?		
6	How many ways can Mrs. Sullivan choose two students from 27 to help put away calculators at the end of class?		
7	You have enough tickets to play 6 different games at the amusement park. If there are 14 games, how many ways can you choose six?		
8	How many different ways can 9 trumpet players in the marching band line up?		
9	Seven students worked together on a project. How many ways can their teacher choose four to present the project?		
10	There are 18 offensive players on the hockey team. How many ways can the coach choose a left wing, center, and right wing to start the game?		
11	How many different 12-letter arrangements can be made using the letters in the word INDIANAPOLIS ?		
12	There are 26 gold fish in the tank at the store. How many ways can Ben choose five?		



Name: _____

Unit 11: Probability & Statistics

Date: _____ Bell: _____

Homework 1: Fundamental Counting Principle,
Permutations & Combinations

**** This is a 2-page document! ****

Directions: Find the total number of outcomes that are possible.

1. In a school building, there are 8 exterior doors and 12 stairways to enter the second floor. Find the total number of ways to reach the second floor.	2. Kendall went shopping and purchased 7 shirts, 3 pairs of pants, 2 jackets. How many different outfits are possible?
3. An internet passcode consists of a digit followed by a letter, followed by another digit. Assuming the digits are 0-9. How many different passcodes are possible?	4. Grant is rolling a standard six-sided number die eight times. How many outcomes are possible?
5. A quiz has eight multiple choice questions with four options for each (A, B, C, and D). How many ways are there to answer the questions?	6. When Jack bought his new truck, there were 96 different ways his truck could be equipped. He had four choices of engines and two choices of transmissions. If the only other choice was color, how many colors were available?

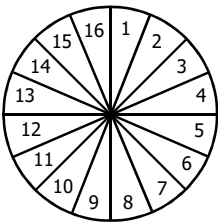
Directions: Evaluate the following.

7. $6!$	8. $\frac{16!}{7!}$	9. $\frac{7!}{10!}$	10. $\frac{2! \cdot 9!}{(10 - 7)!}$
11. ${}_{20}P_4$	12. ${}_{15}C_8$	13. ${}_{17}P_7$	14. ${}_{27}C_{22}$

15. Find ${}_9P_9$ and ${}_9C_9$. Why do the answers differ? Explain.

Directions: Determine whether the situations represent permutations or combinations. Then solve.		
Example	P/C	Solve
16. Scott has 7 chores to complete this Saturday. How many ways can he arrange the order in which he does them?		
17. There are 16 different colored markers in the bucket. How many ways can Kelly choose five of them?		
18. How many ways are there to elect a President, Vice President, Secretary, and Treasurer, from a club with 32 members?		
19. The dance company is choosing 3 new dancers from a group of 25 who try out. How many ways can they choose the new dancers?		
20. There are ten toppings available to make an ice cream sundae. How many ways can Max choose two?		
21. How many different 11-letter arrangements can be made using the letters in the word FIRECRACKER ?		
22. Mindy is purchasing songs from an album on iTunes. If the album has 14 songs, how many ways can she choose 6?		
23. Carl is choosing a four-character password for his cell phone. The password can contain letters and digits only, and can't include repeating characters. How many passwords are possible?		
24. Mr. Pratt has 26 students in his math class. He has three prizes to give away: a pencil, an eraser, and a no homework coupon. How many ways can he choose three students to win these awards?		
25. How many ways are there to choose seven cards from a deck of 52?		

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
Theoretical Probability	- Probability is the measure of how _____ an event is to occur. - The set of all possible outcomes is called the sample space. - For equally likely outcomes, the theoretical probability of an event, $P(E)$, is the _____ of the number of favorable outcomes to the total number of outcomes possible.	
Simple Events The probability of one event.	1. A jar contains 32 red marbles and 28 blue marbles. What is the probability that a randomly selected marble is red?	2. A letter in the word RESTORATION is randomly selected. What is the probability of selecting a vowel?
	3. A day in the month of January is randomly selected. What is the probability of selecting a prime number?	4. Two dice are rolled. What is the probability that the sum of the two dice on the next roll is at least 9?
	5. What is the probability of drawing a heart or a club from a standard deck of cards?	6. There are 8 books lettered A-H on the shelf. If Scott randomly chose two books, what is the probability that he chose books A and B?
Complement of an Event 	The complement of an event is the probability of the event _____ happening. Since the sum of all probabilities in sample space is _____, the probability of an event not happening is $P(\sim E) = \rule{1.5cm}{0.4pt}$.	
	7. The probability that it will snow tomorrow is $\frac{7}{20}$. What is the probability that it will not snow?	8. A month of the year is randomly selected. What is the probability of getting a month that does not begin with the letter A?
	9. If the spinner to the left is spun, find the probability that it lands on a number that is not prime.	10. Two dice are rolled. What is the probability of not getting doubles?

Compound Events The probability of two or more simple events.	Independent Events When the outcome of one event <u>does not affect</u> the outcome of the other event. $P(A \text{ and } B) =$	Dependent Events When the outcome of one event <u>does affect</u> the outcome of the other event. $P(A \text{ and } B) =$
Independent Events	11. A die is rolled 3 times. What is the probability of getting 1's on each roll?	12. A coin is tossed, then a day of the week is selected. What is the probability of getting tails then a day starting with the letter T?
	A bag contains 8 red crayons, 14 purple crayons, 6 yellow crayons, and 4 green crayons. A crayon is selected, replaced, then another is selected. Find each probability.	
	13. $P(\text{purple then yellow})$	14. $P(\text{green then red})$
	15. $P(\text{two purples})$	16. $P(\text{two yellows})$
Dependent Events	Using the same example from above, assume once a crayon is selected, it is NOT replaced. Find each probability.	
	17. $P(\text{yellow then red})$	18. $P(\text{purple then green})$
	19. $P(\text{two reds})$	20. $P(\text{two greens})$
	21. A card is drawn from a standard deck, not replaced, and another is drawn. What is the probability of choosing a heart then a spade?	22. Jack had four Snicker bars and 8 Mars bars. He randomly chose a piece of candy, ate it, then chose another. What is the probability that both candy bars were Snickers?



Name: _____

Unit 11: Probability & Statistics

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Homework 2: Theoretical Probability

**** This is a 2-page document! ****

Part I: Simple Probability

Use for questions 1-3: A random two-digit number (10-99) is drawn. Find each probability.

1. $P(32)$

2. $P(\text{odd number})$

3. $P(\text{a multiple of 5})$

Use for questions 4-6: A letter is randomly chosen from the word **CANDLESTICK**. Find each probability.

4. $P(\text{a vowel})$

5. $P(N \text{ or } S)$

6. $P(\text{not } C)$

7. Three coins are tossed. Find the probability that two land on heads.

8. A month is randomly chosen. What is the probability that the month chosen has less than 31 days?

9. What is the probability of drawing a 9 or diamond from a standard deck of cards?

10. Credit cards place a three-digit security code on the back of cards. What is the probability that a code starts with the number 7?

11. Two dice are rolled. What is the probability of not getting doubles?

12. Mikayla has the following songs on her iPod: 14 Taylor Swift songs, 16 Meghan Trainor songs, and 17 Katy Perry songs. What is the probability that the next song that plays is not Katy Perry?

Part II: Compound Probability

13. A dice is rolled, then a coin is tossed. What is the probability of getting a 5 then tails?

14. A coin is tossed, then a number 1-10 is chosen at random. What is the probability of getting heads then a number less than 4?

15. Natalie guessed on the last four true or false questions on her math quiz. What is the probability that she got all four questions correct?	16. A card is drawn from a standard deck and a letter is chosen from the word INCREDIBLE. What is the probability of drawing a king then getting an I?
Use for questions 17-20: A bag contains 30 lottery balls numbered 1-30. A ball is selected, replaced, then another is drawn. Find each probability.	
17. $P(\text{and even, then odd})$	18. $P(7, \text{ then a number greater than } 16)$
19. $P(\text{a multiple of } 5, \text{ then a prime number})$	20. $P(\text{two even numbers})$
Use for questions 21-24: A bag contains 30 lottery balls numbered 1-30. A ball is selected, NOT replaced, then another is drawn. Find each probability.	
21. $P(\text{a 2-digit number, then } 4)$	22. $P(19, \text{ then a multiple of } 4)$
23. $P(24, \text{ then a number less than } 15)$	24. $P(\text{two perfect squares})$
25. A football team has 5 freshmen, 8 sophomores, 11 juniors, and 16 seniors. If two are chosen at random to participate in the coin toss, what is the probability that both players chosen are seniors?	26. Ryan's mom randomly chooses two days each week for Ryan to do his chores. What is the probability that she picks Saturday and Sunday?

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples										
Conditional Probability	- Given events A and B, conditional probability is the probability of event B occurring, knowing that event A has _____. - Notation: _____ - Read as: _____ Conditional Probability Formula: 										
Examples	1. Mitchell drew a card from a standard deck of playing cards. What is the probability that he drew a queen, given that the card was red?	2. A number from 1-100 is randomly selected. What is the probability that it is a perfect square, given that it is an odd number?									
	3. There are 62 people that take yoga class and 48 people that take spinning class at the gym. Fifteen people that take yoga also take spinning. If a person from these groups is selected at random, find the probability that he or she takes yoga class, if you know they take spinning class.										
	4. Of the 30 students in Mrs. Smith's math class, 13 are athletes and 9 are in band. Three of the athletes are also in band. If one student is chosen at random, find the probability that this student is in band, if it is known that they are not an athlete.										
Two-Way Tables	A table that records data that pertains to two different categories. Example: The table shows the results of a poll of 50 randomly selected students who were asked whether they prefer to watch hockey or football. Find each probability. <table border="1" style="margin-bottom: 10px;"> <thead> <tr> <th></th><th>Boys</th><th>Girls</th></tr> </thead> <tbody> <tr> <th>Prefer Football</th><td>18</td><td>6</td></tr> <tr> <th>Prefer Hockey</th><td>10</td><td>16</td></tr> </tbody> </table> a) $P(\text{prefer football} \mid \text{girl})$ b) $P(\text{boy} \mid \text{hockey})$			Boys	Girls	Prefer Football	18	6	Prefer Hockey	10	16
	Boys	Girls									
Prefer Football	18	6									
Prefer Hockey	10	16									

	Directions: Find each probability using the table.																																
	5. The table shows the results of a poll of 150 randomly selected middle school students who were asked if they take French or Spanish.	a) $P(\text{a 7}^{\text{th}} \text{ grader who takes French})$																															
	<table><tr><td></td><td>French</td><td>Spanish</td></tr><tr><td>6th Grade</td><td>19</td><td>23</td></tr><tr><td>7th Grade</td><td>25</td><td>12</td></tr><tr><td>8th Grade</td><td>38</td><td>33</td></tr></table>		French	Spanish	6 th Grade	19	23	7 th Grade	25	12	8 th Grade	38	33	b) $P(8^{\text{th}} \text{ grader} \mid \text{takes Spanish})$																			
	French	Spanish																															
6 th Grade	19	23																															
7 th Grade	25	12																															
8 th Grade	38	33																															
		c) $P(\text{takes French} \mid 6^{\text{th}} \text{ grader})$																															
		d) $P(7^{\text{th}} \text{ grader} \mid \text{takes French})$																															
	6. The table shows the results of a poll of randomly selected juniors and seniors who were asked if they attended prom.	a) $P(\text{a junior that did not attend prom})$																															
	<table><tr><td></td><td></td><td>Juniors</td><td>Seniors</td></tr><tr><td rowspan="2">Prom</td><td>Yes</td><td>28</td><td>97</td></tr><tr><td>No</td><td>56</td><td>19</td></tr></table>			Juniors	Seniors	Prom	Yes	28	97	No	56	19	b) $P(\text{did not attend prom} \mid \text{senior})$																				
		Juniors	Seniors																														
Prom	Yes	28	97																														
	No	56	19																														
		c) $P(\text{junior} \mid \text{attended prom})$																															
Frequencies	Joint Relative Frequencies: The values in each category divided by total number of values. Marginal Relative Frequencies: Found by totaling each row and column.																																
Examples	7. The table below shows the number of first class and coach class passengers on a plane who checked one bag, two bags, or no bags. Find the joint relative and marginal relative frequencies, then find each probability.																																
	<table><tr><td></td><td>First Class</td><td>Coach Class</td></tr><tr><td>2 bags</td><td>16</td><td>52</td></tr><tr><td>1 bag</td><td>12</td><td>60</td></tr><tr><td>0 bags</td><td>2</td><td>18</td></tr></table>		First Class	Coach Class	2 bags	16	52	1 bag	12	60	0 bags	2	18	<table><tr><td></td><td>First Class</td><td>Coach Class</td><td>Total</td></tr><tr><td>2 bags</td><td></td><td></td><td></td></tr><tr><td>1 bag</td><td></td><td></td><td></td></tr><tr><td>0 bags</td><td></td><td></td><td></td></tr><tr><td>Total</td><td></td><td></td><td></td></tr></table>		First Class	Coach Class	Total	2 bags				1 bag				0 bags				Total		
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1 bag																																	
0 bags																																	
Total																																	
	a) $P(\text{a coach class passenger that did not check any bags})$	b) $P(\text{a first class passenger that checked at least one bag})$																															
	c) $P(\text{coach class} \mid \text{checked one bag})$	d) $P(\text{checked two bags} \mid \text{first class})$																															



Name: _____

Unit 11: Probability & Statistics

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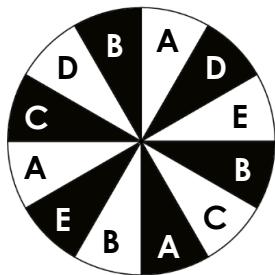
Homework 3: Conditional Probability

**** This is a 2-page document! ****

Use for questions 1-2: A bucket contains 50 lottery balls numbered 1-50. One is drawn at random. Find each probability.

1. $P(\text{multiple of 6} \mid \text{2-digit number})$	2. $P(\text{at least 20} \mid \text{prime number})$
3. Marti rolls two dice. What is the probability that the sum of the dice is 7, given that the first die is showing a 2?	4. Blake randomly chose a letter from alphabet. What is the probability that this letter has at least one line of symmetry, given that it is a consonant?
5. A card is randomly selected from a standard deck of playing cards. Find the probability that it is a face card, given that a black card is drawn.	6. A month of the year is randomly chosen. Find the probability that it has no more than 30 days, given that it starts with the letter A.

Use for 7-9: The wheel below is spun. Find each probability.



7. $P(\text{black} \mid \text{A})$
8. $P(\text{C} \mid \text{white})$
9. $P(\text{black} \mid \text{B or E})$

10. Out of the 125 children at summer camp, 45 signed up for swimming and 38 signed up for arts and crafts. Twelve students who signed up for swimming also signed up for arts and crafts. If a child is randomly selected, what is the probability that they are signed up for swimming, if it is known that they did not sign up for arts and crafts?

11. Out of the 56 players on the football team, 24 are on honor roll and 18 have perfect attendance. Seven who are on honor roll also have perfect attendance. If a player is chosen at random, what is the probability that they are on honor roll, if it is known that they also have perfect attendance?

Directions: Find each probability using the table.

- 12.** The table below shows the number of students that do or do not have their own car and whether they have part time jobs.

		Part Time Job	
		Yes	No
Car	Yes	78	18
	No	30	24

- a)** $P(\text{a student with a part-time job without a car})$

- b)** $P(\text{no car} \mid \text{does not have a part-time job})$

- c)** $P(\text{part-time job} \mid \text{car})$

- 13.** The table below shows the fate of the first, second, and third class passengers on the Titanic.

	Survived	Died
1 st Class	199	120
2 nd Class	117	155
3 rd Class	172	537

- a)** $P(\text{a 3rd class survivor})$

- b)** $P(\text{1st class} \mid \text{died})$

- c)** $P(\text{survived} \mid \text{2nd class})$

- d)** $P(\text{1st or 3rd class} \mid \text{survived})$

- 14.** The table below represents the GPA of a group of undergrad students and whether or not they will be attending grad school. Find the joint relative and marginal relative frequencies.

		Attending Grad School	
		Yes	No
GPA	≥ 3.0	96	18
	< 3.0	32	54

		Attending Grad School		
		Yes	No	Total
GPA	≥ 3.0			
	< 3.0			
Total				

If a student is chosen at random, find each probability:

- a)** $P(\text{not attending grad school})$

- b)** $P(< 3.0 \text{ GPA and not attending grad school})$

- c)** $P(\text{not attending grad school} \mid \geq 3.0 \text{ GPA})$

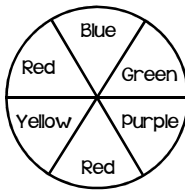
- d)** $P(< 3.0 \text{ GPA} \mid \text{attending grad school})$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
PASCAL'S TRIANGLE	Pascal's triangle was a pattern of numbers that was discovered in the 13th century. Each number in Pascal's triangle is the sum of the two numbers diagonally above it. All outside numbers are 1. Complete rows 4 and 5 of Pascal's triangle below: Row 0 → Row 1 → Row 2 → Row 3 → Row 4 → Row 5 →
<i>Expanding Binomials</i>	Expand the binomial $(a + b)^3$: What do you notice about the coefficients?
THE BINOMIAL THEOREM	If n is a natural number, then $(a + b)^n =$ ${}_nC_0 \cdot a^n b^0 + {}_nC_1 \cdot a^{n-1} b^1 + {}_nC_2 \cdot a^{n-2} b^2 + \dots + {}_nC_n \cdot a^0 b^n = \sum_{k=0}^n {}_nC_k \cdot a^{n-k} b^k$
<i>Examples</i>	Directions: Use the binomial theorem to expand each binomial. 1. $(a + b)^5$ 2. $(x + y)^7$

	3. $(c + d)^{10}$
<i>Coefficients Other than 1</i>	4. $(x - 3)^6$
	5. $(2m + n)^7$
	6. $(k + 2)^8$
	7. $(3p - 2q)^5$
Observations	In the binomial expansion of $(a + b)^n$: • The total number of terms is always _____. • The exponent of a in the first term is _____. • The exponent of b in the last term is _____. • The exponent of a _____ from left to right. • The exponent of b _____ from left to right. • The sum of the exponents in each term is _____. • The coefficients are _____ and follow the _____ row of _____.

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples																
Binomial Experiments	A binomial experiment consists of n independent trials in which: - There are only two outcomes: success and failure - The probability of success p is the same in every trial. - The probability of failure q is the same in every trial. - The probability of success is _____. - The probability of failure is _____.																
Examples	<table><tr><th>Experiment</th><th>Success</th><th>Failure</th><th>p</th><th>q</th></tr><tr><td>Randomly selecting a month 30 times</td><td>January</td><td>Not January</td><td></td><td></td></tr><tr><td>Randomly selecting a card 100 times</td><td>Heart</td><td>Not a Heart</td><td></td><td></td></tr></table>		Experiment	Success	Failure	p	q	Randomly selecting a month 30 times	January	Not January			Randomly selecting a card 100 times	Heart	Not a Heart		
Experiment	Success	Failure	p	q													
Randomly selecting a month 30 times	January	Not January															
Randomly selecting a card 100 times	Heart	Not a Heart															
Binomial Probability	If a binomial experiment has n trials in which p is the probability of success and q is the probability of failure, then the binomial probability that there will be exactly r successes is:																
Examples (with an exact number of successes)	1. Mark tosses a coin 25 times. What is the probability that it lands on heads exactly seven times?	2. The spinner below is spun 10 times. What is the probability that it lands on red exactly six times? 															
	3. Kate took a multiple choice test with four choices for each question. If she guessed the last 5 questions, what is the probability that she got exactly three questions correct?	4. Chad randomly chooses a card from a standard deck 30 times. What is the probability that he gets a diamond exactly four times?															

	5. There are 30 balls numbered 1-30 in a bag. One is randomly selected 12 times. What is the probability of getting a prime number exactly five times?	6. The probability of customers at a store receiving a coupon printed on their receipt is 1 in 8. If 20 customers visit the store today, what is the probability that exactly three of them receive coupons?
	7. In a game of chance, the probability of winning is 30%. If 15 games are played, find the probability of winning exactly six games.	8. A certain traffic light is red for 20 seconds, yellow for 4 seconds, and green for 36 seconds. Find the probability that of the next 12 cars that randomly arrive at the light, exactly 4 will be stopped by a red light.
Examples (with at least, at most, etc.)	9. Kyle tosses a coin 10 times. What is the probability that he gets tails at least 8 times?	10. Carol randomly chooses a day of the week six times. What is the probability of choosing Tuesday or Thursday at most three times?
	11. The soda company prints prizes inside their bottle caps. The probability of winning a prize is 1 in 3. Out of eight bottle caps, what is the probability that at least five are winners?	12. There are 7 boys and 5 girls on the debate team. Each week, one is chosen at random to lead the debate. For the next 10 weeks, what is the probability that no more than four girls are chosen to lead?



Name: _____

Unit 11: Probability & Statistics

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Homework 4: Binomial Theorem & Binomial Probability

**** This is a 2-page document! ****

Directions: Expand each binomial using the binomial theorem.

1. $(a + b)^9$

2. $(m - 2)^8$

3. $(3k - 1)^6$

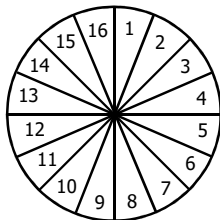
4. $(2r + 5s)^5$

Directions: Find each binomial probability.

5. The probability that Ashley wins a race against Madelyn is 25%. What is the probability that Ashley wins **exactly** three of the next five races against Madelyn?

6. A coin is tossed 14 times. What is the probability of getting heads **exactly** 5 times?

7. The spinner below is divided into 16 equally spaced sections. If the spinner is spun 20 times, what is the probability of getting a perfect square **exactly** seven times?



8. The probability of hitting a target is $\frac{1}{8}$. What is the probability of hitting the target **exactly** twice in five tries?

9. A die is rolled eight times. What is the probability of getting **exactly** three 6's?

10. There are five blue crayons, seven yellow crayons, and eight red crayons in a box. If one is randomly drawn and replaced 15 times, find the probability of drawing **exactly** four blue crayons.

11. In hockey, Ed makes 7 goals for every 10 shots. If he takes 6 shots, what is the probability that he will make **at least** 5 goals?

12. The probability that it will snow on the last day of January is 85%. If the probability remains the same for the first eight days of February, what is the probability that it will snow **at least** five of those days in February?

13. Whenever Nicole rents a movie from iTunes, the probability that it will be a comedy is 52%. Of the next seven movies she rents, what is the probability that she rents **no more than** two comedies?

14. The chance of winning a certain game at a carnival is 2 in 5. If Andy plays the game 12 times, what is the probability that he loses **at most** 3 times?

Name: _____

Algebra 2

Date: _____ Per: _____

Unit 11: Probability & Statistics

Quiz 11-1: Probability

Part I – Counting Principle, Permutations, and Combinations

1. A six-sided number cube is rolled and a card is drawn from a standard deck. Find the total number of outcomes. 1. _____
2. A science quiz has eight multiple choice questions with five choices for each. Find the total number of ways to answer the questions. 2. _____
3. The golf club can send a team of four students to the next golf tournament. If there are 15 students in the club, how many different teams of four can be selected? 3. _____
4. There are 18 skaters competing in the competition. How many ways can they win the gold, silver, and bronze medal? 4. _____
5. What is the total number of different 10-letter arrangements that can be formed using the letters in the word **SUSPICIOUS**? 5. _____

Part II – Probability: Give each answer as a fraction in simplest form.

6. What is the probability of drawing a jack or a red card from a standard deck of playing cards? 6. _____
7. A bag contains 40 slips of paper numbered 1-40. If one is randomly chosen, what is the probability of not choosing a multiple of 3? 7. _____
8. A coin is tossed two times, then a month of the year is randomly selected. What is the probability of getting tails each time, and a month that starts with the letter J? 8. _____
9. A bookshelf contains 9 mysteries, 7 biographies, and 4 science fiction books. A book is randomly selected, not replaced, then another is selected. Find the probability of selecting a biography then a mystery. 9. _____
10. A shape is randomly selected from the following quadrilaterals: parallelogram, rhombus, rectangle, square, and trapezoid. What is the probability that it has four right angles, given that it has four congruent sides? 10. _____

- 11.** Of the 100 students enrolled at the dance studio, 50 take ballet and 24 take tap dancing. Eight of the students who take ballet also take tap dancing. If a student at the studio is chosen at random, find the probability that they take ballet, if it is known that they do not take tap dancing. **11.** _____

- 12.** The table below represents the number of NFL teams that won or lost their first game of the season. Find the joint relative and marginal relative frequencies.

	Conference	
	AFC	NFC
Won	10	6
Lost	6	10

	Conference		
	AFC	NFC	Total
Won			
Lost			
Total			

Use the tables above to find each probability.

- 13.** $P(\text{either AFC or NFC won})$ **13.** _____

- 14.** $P(\text{NFC} \mid \text{won})$ **14.** _____

- 15.** $P(\text{lost} \mid \text{AFC})$ **15.** _____

Part III – Binomial Probability: Give each answer as a percent rounded to the nearest hundredth.

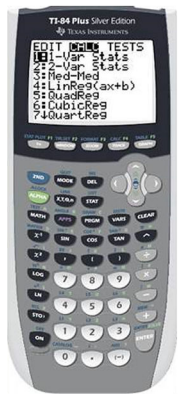
- 16.** A coin is tossed 12 times. What is the probability of getting heads exactly 7 times? **16.** _____

- 17.** Northern Airlines has a reputation for being on time 86% of the time. Find the probability that a Northern Airlines plane will be on time for at least four of its next five flights. **17.** _____

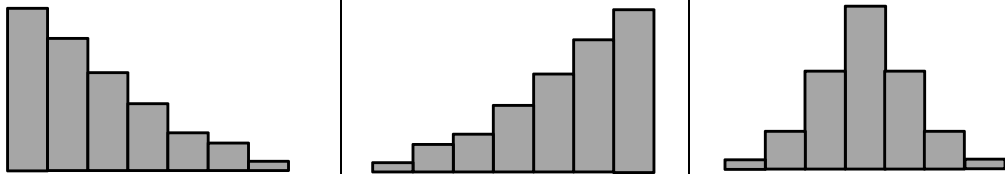
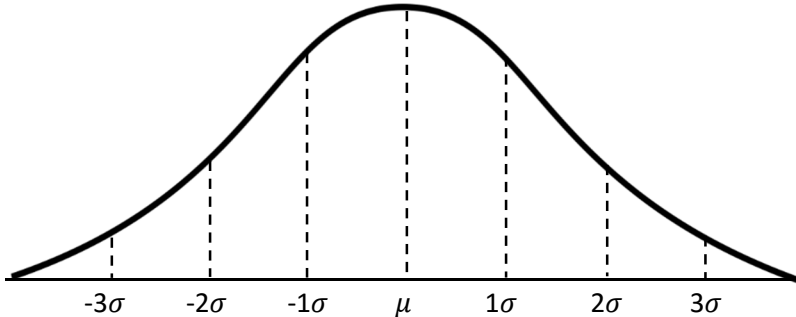
- 18.** Scott strikes out at bat 36% of the time. Find the probability that he strikes out no more than three times on his next eight times at bat. **18.** _____

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
MEASURES OF CENTRAL TENDENCY		
	Mean (____): _____	
	Median: _____	
	Mode(s): _____	
	Directions: Find the mean, median, and mode(s) for each data set.	
	1. {58, 53, 59, 51, 46, 35, 51, 58, 60}	
	Mean = _____	
	Median = _____	
	Mode(s) = _____	
2. {21, 10, 27, 24, 15, 7, 19, 24, 31, 15, 11, 24}	Mean = _____	
	Median = _____	
	Mode(s) = _____	
MEASURES OF VARIATION		
Mean Absolute Deviation (MAD)		$MAD = \frac{\sum_{i=1}^n x_i - \mu }{n}$
	Directions: Find the mean absolute deviation for each data set.	
	3. {85, 74, 88, 80, 92, 60}	
	MAD = _____	
4. {14, 18, 16, 19, 21, 14, 15, 23, 21, 19}		
	MAD = _____	

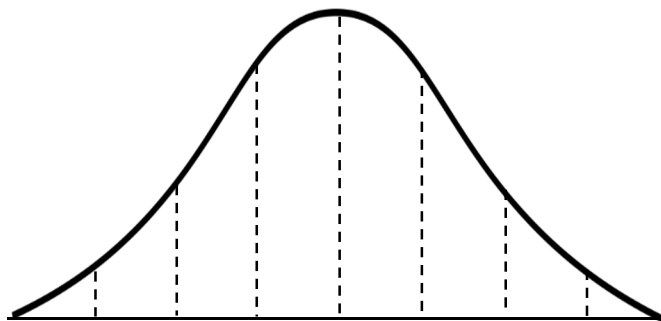
Variance (σ^2)		$\sigma^2 = \frac{\sum_{i=1}^n (x_i - \mu)^2}{n}$
Standard Deviation (σ)		$\sigma = \sqrt{\frac{\sum_{i=1}^n (x_i - \mu)^2}{n}}$
What do the numbers mean?	• A lower standard deviation indicates that _____ • A higher standard deviation indicates that _____	
Examples	Directions: Find the variance and standard deviation for each data set. 5. {85, 88, 91, 96, 100} Variance (σ^2) = _____ Standard Deviation (σ) = _____	
Calculator Steps: • Hit STAT, ENTER • Enter the data values in L₁ • Hit STAT • Arrow over to CALC • Choose #1:1-Var Stats • Hit ENTER • Scroll down to find σ • Square to find σ^2 	6. {3, 4, 7, 8, 11, 12, 12, 15, 19} Variance (σ^2) = _____ Standard Deviation (σ) = _____	
	7. {52, 47, 60, 55, 58, 53, 59, 63} Variance (σ^2) = _____ Standard Deviation (σ) = _____	
	8. {127, 81, 135, 100, 97, 115, 72, 152, 63, 164} Variance (σ^2) = _____ Standard Deviation (σ) = _____	
	9. {25, 27, 21, 32, 35, 29, 23, 26, 24, 25, 28, 31} Variance (σ^2) = _____ Standard Deviation (σ) = _____	

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
Data Distribution	
Normal Distribution	- A symmetrical _____ defined by the mean (μ) and the standard deviation (σ) of the data set. - The area under the curve represents _____, with the total area equal to _____.
The Empirical Rule	In a normal distribution with mean μ and standard deviation σ: - Approximately _____ of the data falls within 1σ of the mean. - Approximately _____ of the data falls within 2σ of the mean. - Approximately _____ of the data falls within 3σ of the mean. 

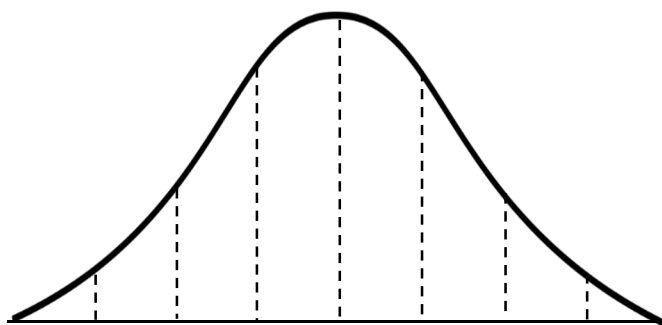
Directions: Draw and label normal distribution curves, then answer the questions.

1. The weights of the 50 football players are normally distributed with a mean of 178 pounds and a standard deviation of 8 pounds.



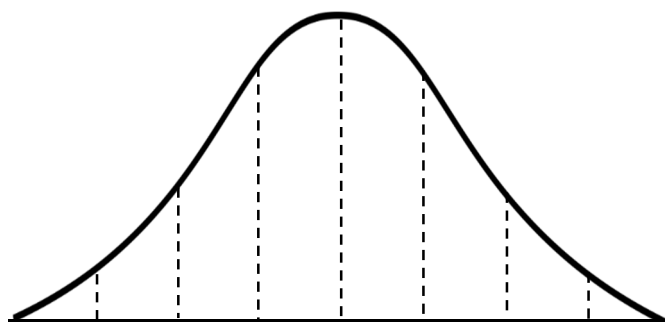
- What percent of the players weigh between 178 lbs and 194 lbs?
- What is the probability that a player weighs at most 170 lbs?
- What is the probability that a player weighs less than 162 lbs or greater than 194 lbs?
- How many players weigh between 170 lbs and 186 lbs?

2. A set of 120 test scores are normally distributed with a mean of 82 and a standard deviation of 5.



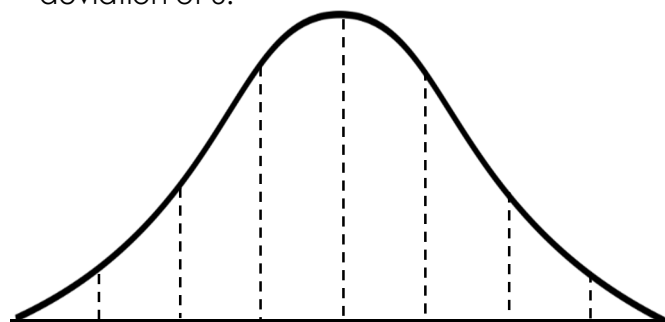
- What percent of the scores are between 72 and 87?
- What is the probability that a score is greater than 77?
- What is the probability that a score is less than 82 or greater than 92?
- About how many students scored outside two standard deviations of the mean?

3. The price of a gallon of regular gasoline at 75 gas stations across the state is normally distributed with a mean of \$2.05 and a standard deviation of 4¢.



- What percent of gas stations sell a gallon of regular gas for less than \$1.97?
- What percent of gas stations sell a gallon of regular gas for at least \$2.17?
- What is the probability that a gas station sells a gallon of regular gas for less than \$1.97 or greater than \$2.05?
- About how many stations sell a gallon of regular gas for no more than \$2.01?

4. Mrs. Fuller recently tested her 120 keyboarding students to see how many words per minute they can type. The results were normally distributed with a mean of 45 and a standard deviation of 6.



- About how many students can type at least 39 words per minute?
- About how many students can type within one standard deviation of the mean?
- Students need to be in the top 2% in order to be eligible for the national typing competition. If Carla can type 56 wpm, is she eligible?



Name: _____

Unit 11: Probability & Statistics

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Homework 5: Measures of Center, Variation, and Normal Distribution

**** This is a 2-page document! ****

Directions: Find the mean, median, and mode of each data set below.

1. {11, 14, 11, 5, 17, 28, 3}

Mean = _____
Median = _____
Mode(s) = _____

2. {24, 29, 31, 16, 49, 52, 29, 35, 62, 29}

Mean = _____
Median = _____
Mode(s) = _____

Directions: Find the mean absolute deviation, variance, and standard deviation for each data set.

3. The following data shows the number of fish caught by seven boy scouts on their camping trip:
{1, 2, 2, 4, 5, 6, 8}

MAD = _____
Variance: σ^2 = _____
Standard Deviation: σ = _____

4. The following data shows the points scored by a football team during their first ten games of the season: {27, 32, 41, 9, 14, 20, 31, 33, 20, 13}

MAD = _____
Variance: σ^2 = _____
Standard Deviation: σ = _____

5. The following data shows the high temperature for the past eight days:
{66, 46, 53, 50, 52, 47, 45, 49}

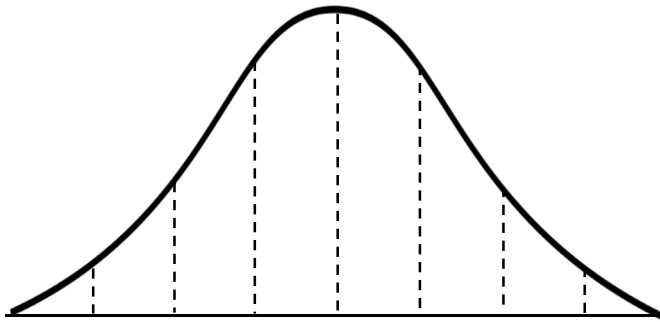
MAD = _____
Variance: σ^2 = _____
Standard Deviation: σ = _____

6. The following data shows the price of six different jars of pasta sauce at the grocery store:
{\$2.79, \$1.99, \$4.29, \$2.49, \$2.29, \$3.49}

MAD = _____
Variance: σ^2 = _____
Standard Deviation: σ = _____

For questions 7 and 8, draw the normal distribution curve, then answer the questions.

- 7.** A set of 125 golf scores are normally distributed with a mean of 76 and a standard deviation of 3.

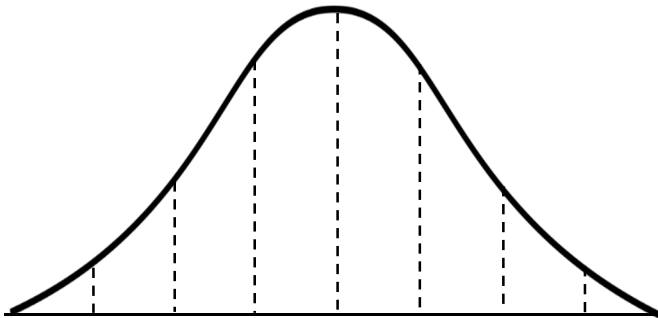


- a)** What percent of the scores are between 67 and 85?

- b)** What is the probability that a score is no more than 79?

- c)** About how many scores fell between one standard deviation of the mean?

- 8.** The talk-time battery life of a group of cell phones is normally distributed with a mean of 5 hours and a standard deviation of 15 minutes.



- a)** What percent of the phones have a battery life of at least 4 hours and 45 minutes?

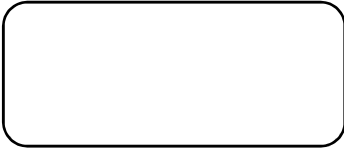
- b)** What percent of the phones have a battery life between 4.5 hours and 5.25 hours?

- c)** What percent of the phones have a battery life less than 5 hours or greater than 5.5 hours?

- 9.** The number of hours that the employees at the grocery store worked last week is normally distributed with a mean of 24 and a standard deviation of 6. If there are 60 total employees, approximately how many worked at least 30 hours last week?

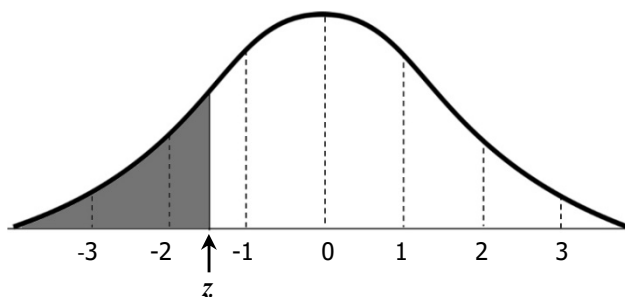
- 10.** The grade point average (GPA) of the students at Lakeview High School is normally distributed with a mean of 3.1 and a standard deviation of 0.3. If there are 1800 students enrolled at the school, approximately how many have a GPA between 2.5 and 3.7?

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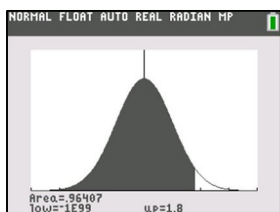
Main Ideas/Questions	Notes/Examples
Z-SCORES	- A value that shows how many _____ a data value is from the _____. - When the z-score is positive, the data value is _____ the mean. - When the z-score is negative, the data value is _____ the mean. - When z = 0, the data value _____ the mean
Z-Score Formula	To find the z-score for a data value in a set that is normally distributed, use:  X = the data value μ = the mean σ = the standard deviation
EXAMPLES	- A radar detector records the speeds of a group of cars that pass by. If the mean is 46 mph and the standard deviation is 2.8 mph, find the z-scores for each data value. a) 52 b) 42 c) 47 - The mean number of total miles ran last week by each member on the track team was 4 with a standard deviation of 1.2. If Clay's z-score was 2.5, how many miles did he run? - A set of test papers is normally distributed with a standard deviation of 5. If Riley scored an 83 with a z-score of -0.4, what was the mean? - The heights of a group of trees is normally distributed with a mean of 14.3 feet. If the z-score for a 20-foot tall tree is 1.9, what is the standard deviation?

Standard Normal Distribution

- A normal distribution in which $\mu = \underline{\hspace{2cm}}$ and $\sigma = \underline{\hspace{2cm}}$.
- The distribution is "standardized" by the $\underline{\hspace{2cm}} - \underline{\hspace{2cm}}$.



Calculator Directions



First, change your window dimensions:

Xmin: -4, Xmax: 4, Ymin: 0, Ymax: 0.5

To graph the curve and find the probability:

- Step 1:** Find the z-score for each data value.
- Step 2:** Hit 2nd -> VARS
- Step 3:** Arrow over to **DRAW**
- Step 4:** Select 1:ShadeNorm(
- Step 5:** Enter the **lower and upper bounds** for the z-scores
- Step 6:** Scroll down and hit **DRAW**

Example: The golf scores in a tournament is normally distributed with a mean of 76 and a standard deviation of 2.6. Find each probability.

5. $P(z < 2.1)$

6. $P(X < 74)$

7. $P(z > -1.5)$

8. $P(X > 81)$

9. $P(-0.2 < z < 1.8)$

10. $P(77 < X < 83)$



Name: _____

Unit 11: Probability & Statistics

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Homework 6: Z-Scores & Standard Normal Distribution

**** This is a 2-page document! ****

1. The number of calories burned at the gym is normally distributed with a mean of 425 and a standard deviation of 51. Find the z -scores for each data value.

a) 268

b) 512

c) 450

2. The air pressure of each tire in a group is normally distributed with a mean of 39.2 pounds per square inch (psi) and a standard deviation of 2.5 psi. If a tire in this group has a z -score of -1.8, find its air pressure.

3. A set of math tests is normally distributed with a mean of 81 and a standard deviation of 5. If Adam's z -score was -0.6 and Leah's z -score was 2.2, how many points higher did Leah score than Adam?

4. The prices of various cereal boxes is normally distributed with a standard deviation of 0.5. If a box of Lucky Charms sells for \$4.29 and has a z -score of 1.6, what is the mean price of a box of cereal?

5. The number of homes sold each year by a realtor is normally distributed with a mean of 54. If the realtor sold 36 homes last year with a z -score of -2.4, what is the standard deviation?

6. The last math exam of the year was normally distributed with a mean of 85 and a standard deviation of 5. Xavier needed at least a 95 on this test to get an A for the year. His teacher gave him his z -score: 1.8. Did Xavier get an A for the year?

7. The 400-meter race times recorded in the boys track meet was normally distributed with a standard deviation of 3 seconds. If David finished the race in 51.2 seconds with a z -score of -2.6, what was the mean time?

Use for questions 8-14: The weights of dogs in a dog show is normally distributed with a mean of 58 pounds and a standard deviation of 17.2 pounds. Use a standard normal distribution curve to find each probability.

8. $P(z < -0.8)$

9. $P(z > 1.32)$

10. $P(-2.8 < z < -1.17)$

11. $P(X > 51)$

12. $P(35 < X < 62)$

13. $P(X < 102)$

14. $P(X > 85)$

15. The high temperatures in Virginia Beach during the month of July are normally distributed with a mean of 83° and a standard deviation of 6° . Find the probability that on a given day during the month, the temperature is greater than 92° .

16. The mean number of hours that the average person watches television each day is 4.18 hours with a standard deviation of 1.19 hours. Find the probability that someone watches between 3 and 5 hours per day.

17. The mean song length on Jack's iPad is 3 minutes and 3 seconds, with a standard deviation of 24 seconds. Find the probability that the next song to play is no more than 3 minutes and 45 seconds.

Name: _____

Algebra 2

Date: _____ Per: _____

Unit 11: Probability & Statistics

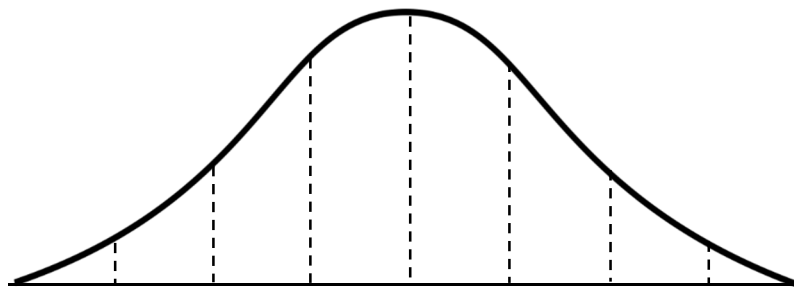
Quiz 11-2: Statistics

1. The following data represents the number of calories in seven sandwiches at the restaurant: {740, 610, 490, 370, 500, 540, 460}. Find the mean absolute deviation. Round your answer to the nearest hundredth. 1. _____
2. The speed in miles per hour of each of Sam's last ten pitches were recorded as follows: {82, 78, 83, 80, 75, 82, 79, 80, 83, 81}. Find the variance. Round your answer to the nearest hundredth. 2. _____
3. The following data below represents the running times of six movies. Find the standard deviation of the running times. Give your answer to the nearest hundredth of a minute. 3. _____

Movie	A	B	C	D	E	F
Running Time	1 hr 47 min	1 hr 32 min	2 hr 14 min	1 hr 56 min	2 hr 32 min	3 hr 2 min

4. The gas mileage for a group of cars is normally distributed with a mean of 24 miles per gallon and a variance of 34.81. What is the standard deviation? 4. _____

Use for questions 5-8: The number of days a group of 250 homes is on the market is normally distributed with a mean of 50 and a standard deviation of 13. Label the normal distribution curve, then answer the questions.



5. What percent of the homes are on the market between 11 and 89 days? 5. _____
6. What is the probability that a home is on the market for at least 37 days? 6. _____
7. Approximately how many homes were on the market between 37 and 63 days? 7. _____
8. Approximately how many homes were on the market for no more than 24 days? 8. _____

9. The number of text messages that teenagers send per month is normally distributed with a mean of 3,400 and a standard deviation of 450. If Kendra sent 4,415 text messages last month, find her z -score to the nearest hundredth. **9.** _____

10. The daily high temperatures in Philadelphia for the first 15 days in February is normally distributed with a standard deviation of 2.5° . If it was 37° on Valentine's Day with a z -score of 2.4, find the mean temperature. **10.** _____

11. The price of a gallon of milk at various stores is normally distributed with a mean of \$2.89 and a standard deviation of \$0.2. The price of a gallon of milk at SmartShop has a z -score of -1.3 while the price of a gallon of milk at Wegmans has a z -score of 0.6. How much more is a gallon of milk at Wegmans compared to SmartShop? **11.** _____

12. Mark recently took a road trip across the country. The number of miles he drove each day was normally distributed with a mean of 450. If he drove 431.8 miles on the last day with a z -score of -0.7, what is the standard deviation? **12.** _____

Use for questions 13-16: The number of points scored by each team in the NHL at the end of the season is normally distributed with a mean of 89 and a standard deviation of 11. Find each probability. (X represents a data value and z represents a z -score)

13. $P(z > 1.3)$ **13.** _____

14. $P(X < 85)$ **14.** _____

15. $P(71 < X < 96)$ **15.** _____

16. If there are 30 teams, approximately how many scored at least 76 points? **16.** _____

PROBABILITY & STATISTICS *Formulas Reference*

Term	Definition	Formula
Fundamental Counting Principle		
Factorial		
Permutation		
Combination		
Independent Events		
Dependent Events		
Conditional Probability		
Binomial Probability		

Binomial
Theorem

Mean
Absolute
Deviation

Variance

Standard
Deviation

z-score

Extra Notes:

Unit 11 Test Study Guide

(Probability & Statistics)

Name: _____

Date: _____ Block: _____

Topic 1: Fundamental Counting Principle, Permutations, & Combinations

1. There are ten questions on Jack's science quiz: six multiple choice questions with four choices each and four true/false questions. How many ways can he answer the questions?	2. The Sandwich Shop offers combo meals with a choice of one sandwich, one side, and one drink. In total, there are 560 combos available. If there are 16 sandwiches and 5 drinks available, how many sides are there?
Determine whether the situation represents a permutation or combination, then solve.	
3. Marlena must answer three out of eight essay questions on her writing test. How many ways can she choose 3?	4. The Grandview High School Band plans to play six pieces of music in their upcoming concert. How many ways can they arrange the order of their performance?
5. How many ways can a president, vice president, and treasurer be elected from a group of 40 students?	6. Josh must submit 4 paintings as part of his application to art school. If he has 25 to choose from, how many ways can he pick 4?
7. How many different 12-letter arrangements can be made using the letters in the word CARELESSNESS?	8. The field hockey coach chooses players to put away equipment after each practice. If there are 16 players on the team, how many ways can she choose five?

Topic 2: Theoretical Probability/Independent vs. Dependent Events

9. Kate is randomly choosing a date in the month of August. What is the probability that she chooses a two-digit number or a prime number?	10. Erik is randomly choosing a card from a standard deck. What is the probability that he chooses a card that is red and a multiple of three?
11. One of the 50 states is chosen at random. What is the probability of not choosing a state that begins with the letter N?	12. Tony's wallet contains six \$1 bills, four \$5 bills, two \$10 bills, and three \$20 bills. If Tony randomly removes a bill, what is the probability that it is at least \$5?

13. A number 1-30 is randomly selected, followed by a letter from the word TARGET. What is the probability of choosing a perfect square then the letter T?	14. Carter is handing out Valentine's Day cards to his friends. He has 8 Batman cards, 12 Superman cards, and 5 Captain America cards in a bag. What is the probability that the first two he chooses are Batman?
Use for questions 15-16: A bag contains 25 red chips, 18 blue chips, and 29 yellow chips. A chip is selected, replaced, then another chip is chosen. Find each probability.	
15. $P(\text{blue then red})$	16. $P(\text{yellow then blue})$
Use for questions 17-18: A box of pizza contains three cheese slices, seven pepperoni slices, and five supreme slices. Mark randomly chooses a slice, eats it, then chooses another. Find each probability.	
17. $P(\text{cheese then supreme})$	18. $P(\text{two pepperonis})$

Topic 3: Conditional Probability

19. Mason randomly chooses a card from a standard deck of playing cards. What is the probability that it is not an ace, given it is a red card?	20. One of the 50 United States is randomly selected. What is the probability that it has two syllables, given it starts with the letter M?									
21. Of the 318 sophomores, 140 are taking Algebra 2 and 102 are taking Chemistry. Twenty-six of those taking Algebra 2 are also taking chemistry. If a sophomore is chosen at random, find the probability that they are taking Algebra 2, if it is known that they do not take Chemistry.										
22. A group of cell phone owners were asked whether they had an Android or apple phone. Use the results in the table to find each probability. <table><tr><td></td><td>Male</td><td>Female</td></tr><tr><td>Android</td><td>34</td><td>19</td></tr><tr><td>Apple</td><td>30</td><td>42</td></tr></table>		Male	Female	Android	34	19	Apple	30	42	a) $P(\text{a male that owns an android})$
		Male	Female							
	Android	34	19							
Apple	30	42								
b) $P(\text{apple} \mid \text{male})$										
c) $P(\text{female} \mid \text{android})$										

- 23.** The table below shows the number of gold, silver, and bronze medals won by the United States and China during the 2012 summer Olympics. Find the joint relative and marginal relative frequencies

	U.S.	China
Gold	46	38
Silver	28	28
Bronze	29	22

	U.S.	China	Total
Gold			
Silver			
Bronze			
Total			

If a medal from those above is randomly selected, find each probability.

a) $P(\text{a U.S. bronze medal})$

b) $P(\text{a gold or silver medal})$

c) $P(\text{silver} \mid \text{China})$

d) $P(\text{U.S.} \mid \text{gold})$

e) $P(\text{China} \mid \text{not silver})$

f) $P(\text{bronze} \mid \text{not China})$

Topic 4: Binomial Probability & Binomial Theorem

Use for questions 24-27: The soda company is printing prizes on the inside of their bottle caps, with a 3 in 5 chance of winning. If Tom purchases a 12-pack of soda, find each probability.

24. $P(\text{exactly two prizes})$

25. $P(\text{exactly five prizes})$

26. $P(\text{at least ten prizes})$

27. $P(\text{no more than three prizes})$

Expand each binomial using the binomial theorem.

28. $(a - 2)^9$

29. $(4k + 1)^5$

Topic 5: Statistics: Measures of Variation, Normal Distribution, and z-Scores

Find the mean absolute deviation, variance, and standard deviation for each data set.

- 30.** The following data represents the last ten field goal attempts, in yards, made by the kicker.
{54, 29, 33, 62, 45, 39, 41, 52, 59, 36}

MAD = _____

Variance: σ^2 = _____

Standard Deviation: σ = _____

- 31.** The following data shows the grades of the Carl's last 8 quiz grades in Algebra 2.
{72, 93, 52, 86, 100, 81, 79, 61}

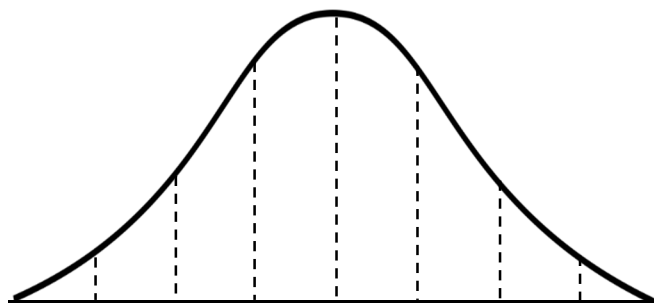
MAD = _____

Variance: σ^2 = _____

Standard Deviation: σ = _____

Label the normal distribution curve, then answer the questions that follow.

The ages of the 32 recruits in police academy are normally distributed with a mean of 27 with a standard deviation of 2.



- 32.** What percent of the recruits are between ages 23 and 27?

- 33.** What is the probability that a recruit is at least 31 years old?

- 34.** Approximately how many recruits are at most 29 years old?

35. The blood pressure of a group of adults is normally distributed with a mean of 122 and a standard deviation of 13. If Mina's blood pressure is 108, find her z-score.	36. The daily balance of a checking account over the course of a month is normally distributed with a standard deviation of \$84.10. If the balance in the account on a particular day was \$317.34 with a z-score of -2.4, find the mean balance.
37. The length of a certain fish species is normally distributed with a mean of 15 cm. If a fish in this species is 18.8 cm with a z-score of 1.9, what is the standard deviation?	38. The number of visits to the gym each year by the members at InFit is normally distributed with a mean of 97 and a standard deviation of 25. If Kurt's z-score is -1.8 and Mindy's z-score is 2.4, how many more times did Mindy visit the gym than Kurt?
Use for questions 39-42: The shoe sizes of the 36 students in Samantha's PE class are normally distributed with a mean of 8.5 and a standard deviation of 1.5.	
39. What percent of the students have a shoe size between 7 and 11?	40. What is the probability that a student will have a maximum shoe size of 9.5?
41. Approximately how many students wear at least a size 6?	42. Approximately how many students wear a shoe size between 8 and 10?

Name: _____

Unit 11 Test

Date: _____ Bell: _____

Probability & Statistics

1. The library labels each book with a five-digit code to identify the book in their computer system. Each code is 3 digits (0-9) followed by two letters. How many codes are possible? A. 468 B. 492,804 C. 527,000 D. 676,000	2. A test has 30 multiple choice questions with 4 choices each. Which expressions represents the total number of ways to answer the questions? A. 4^{30} B. 30^4 C. ${}_{30}P_4$ D. $4 \cdot 30$
3. A coach needs to choose two players from the hockey team to be captain and assistant captain. If there are 20 players on the team, how many ways he can assign the two roles? A. 100 B. 190 C. 380 D. 450	4. In a card game, each player is dealt a hand of 7 cards from a deck of 52 playing cards. Which expression represents the number of hands possible? A. $7! \cdot 52!$ B. $\frac{52!}{7!}$ C. ${}_{52}P_7$ D. ${}_{52}C_7$
5. SweetTooth Frozen Yogurt has 12 different toppings to choose from. If they add two new toppings, how many more ways can Ian choose three? A. 220 B. 144 C. 364 D. 864	6. How many different ways can the 11 letters in the word PROBABILITY be arranged? A. 1,663,200 B. 7,241,600 C. 9,979,200 D. 39,916,800
7. A month of the year is selected at random. What is the probability that it has 31 days or starts with the letter J? A. $\frac{1}{6}$ B. $\frac{3}{4}$ C. $\frac{7}{12}$ D. $\frac{2}{3}$	8. A card is selected at random from a standard deck of 52 playing cards. What is the probability that it is not a face card? A. $\frac{3}{13}$ B. $\frac{10}{13}$ C. $\frac{4}{13}$ D. $\frac{9}{13}$

9. A die is rolled then a letter in the word STATISTICS is randomly selected. What is the probability of at least a three, then the letter S?

A. $\frac{1}{5}$

C. $\frac{3}{20}$

B. $\frac{1}{4}$

D. $\frac{29}{30}$

10. Gavin and Jack are practicing shots against their goalie. On their last 15 attempts, Gavin made 6 and Jack made 7. Based on this performance, what is the probability that they both make their next shot?

A. $\frac{1}{15}$

C. $\frac{13}{15}$

B. $\frac{13}{30}$

D. $\frac{14}{75}$

Use for questions 11-12: A basket contains 8 green apples, 11 red apples, 5 nectarines, and 12 oranges. Mark randomly chooses a piece of fruit, eats it, then selects another. Find each probability.

11. $P(\text{a green apple then a nectarine})$

A. $\frac{23}{63}$

C. $\frac{13}{36}$

B. $\frac{5}{162}$

D. $\frac{2}{63}$

12. $P(\text{two oranges})$

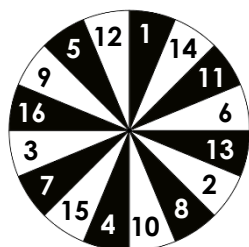
A. $\frac{1}{9}$

C. $\frac{11}{105}$

B. $\frac{4}{35}$

D. $\frac{68}{105}$

13. The spinner below is spun. Find the probability that it lands on an odd number, given that it lands on a white space. Write your answer as a fraction in simplest form.



14. Two dice are rolled. What is the probability that the sum of the dice is at least 5, given that one die is showing a 2.

A. $\frac{1}{2}$

C. $\frac{1}{6}$

B. $\frac{2}{3}$

D. $\frac{1}{3}$

Use for questions 15-16. Out of a group of 120 students that were surveyed about winter sports, 28 said they ski and 52 said they snowboard. Sixteen of the students who said they ski said they also snowboard. If a student is chosen at random, find each probability.

15. $P(\text{snowboard} \mid \text{ski})$

A. $\frac{4}{23}$

C. $\frac{4}{13}$

B. $\frac{4}{17}$

D. $\frac{4}{7}$

16. $P(\text{ski} \mid \text{does not snowboard})$

A. $\frac{3}{17}$

C. $\frac{3}{7}$

B. $\frac{7}{17}$

D. $\frac{3}{13}$

Use for questions 17-18. The table below shows the number of students at Parkland High School that made honor roll last quarter. If a student is chosen at random, find each probability. Give your answers as fractions in simplest form.

	Honor Roll	
	Yes	No
9 th Grade	72	120
10 th Grade	95	117
11 th Grade	76	124
12 th Grade	57	139

17. $P(\text{honor roll} \mid 11^{\text{th}} \text{ grader})$

18. $P(9^{\text{th}} \text{ grader} \mid \text{not honor roll})$

Use for questions 19-20. Rob and Casey are at batting practice. The table below shows the joint relative frequencies for their misses, fouls, hits, and home runs. Give the marginal relative frequencies, then find each probability.

	Rob	Casey	Total
Misses	.05	.1125	
Fouls	.10	.05	
Hits	.2875	.3375	
Home runs	.0375	.025	
Total			

19. $P(\text{home run} \mid \text{Casey})$

A. $\frac{1}{40}$ B. $\frac{2}{5}$ C. $\frac{1}{21}$ D. $\frac{2}{15}$

20. $P(\text{Rob} \mid \text{hit})$

A. $\frac{15}{32}$ B. $\frac{23}{80}$ C. $\frac{23}{50}$ D. $\frac{23}{38}$

21. The odds of winning a game at the carnival are 3 in 7. If Liza plays the game nine times, what is the probability that she wins exactly two times?

- A. 13.15%
B. 3.12%
C. 10.84%
D. 8.23%

22. When Sophia's mom picks her up from school, she is late one third of the time. For the next 15 days, what is the probability that she will be late at most three days?

- A. 79.08%
B. 79.31%
C. 20.69%
D. 20.92%

23. When Tom plays darts, he hits the target 65% of the time. Find the probability that he hits the target at least four out of next six attempts.

- A. 57.17%
B. 64.71%
C. 42.83%
D. 35.29%

24. What is the fourth term in the binomial expansion of $(2x - y)^7$?

- A. $280x^3y^4$
B. $560x^4y^3$
C. $-280x^3y^4$
D. $-560x^4y^3$

Use for questions 25-26: The following data represents the high temperatures for the next several days in Buffalo, NY: {17, 26, 3, 13, 29, 34, 32, 30, 41}

25. Find the mean absolute deviation. Round your answer to the nearest hundredth of a degree.

26. Which of the following values represent the standard deviation and variance for the data?

- A.** Standard Deviation = 9.33; Variance = 3.05
- B.** Standard Deviation = 9.33; Variance = 87.05
- C.** Standard Deviation = 11.16; Variance = 124.55
- D.** Standard Deviation = 11.16; Variance = 3.34

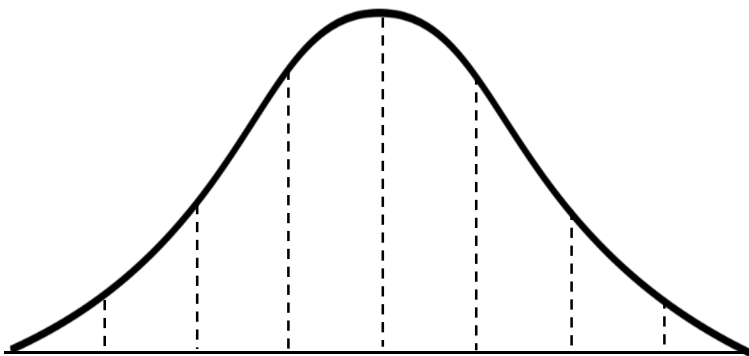
27. Given Data Set A : {12, 14, 17, 5, 10, 8}
If 6 is added to this data set, which statement accurately describes the change in the disbursement of data.

- A.** It will decrease.
- B.** It will increase.
- C.** It will stay the same.
- D.** It's impossible to tell.

28. If the mean and standard deviation of a data set are given, which data set has a variance equal to 16?

- A.** Mean = 4; Standard Deviation = 8
- B.** Mean = 256; Standard Deviation = 8
- C.** Mean = 4; Standard Deviation = 256
- D.** Mean = 256; Standard Deviation = 4

The wind speeds throughout the month of April were normally distributed with a mean of 17 mph and a standard deviation of 4 mph. Label the normal distribution curve, then answer questions 29-32:



29. What percent of the days had wind speeds between 5 and 13 mph?

30. What percent of the days had wind speeds of at least 9 mph?

31. What is the probability that a day had a wind speed of no more than 21 mph?

32. Approximately how many days had wind speeds between 17 and 25 mph?

33. The time it takes to complete a test is normally distributed with a mean of 45 minutes and a standard deviation of 12 minutes. If Nick took an hour and 8 minutes to complete the test, find his z-score to the nearest hundredth.	34. The mean price of a computer sold at a particular store is \$949 with a standard deviation of \$75. The z-score of a certain computer is -0.8. What is the price of the computer?
35. The number of ounces of water a person drinks per day is normally distributed with a standard deviation of 15 ounces. If Sean drinks 88 ounces per day with a z-score of 1.6, what is the mean ounces of water a day that a person drinks?	36. The scores on an Algebra 2 midterm were normally distributed with a mean of 82. If Kendall scored 67 with a z-score of -2.5, what is the standard deviation?
Use for questions 37-40: The 100-meter dash times in the girls track meet were normally distributed with a mean of 13 seconds and a standard deviation of 0.3 seconds.	
37. What is the probability that a runner finished between 12.4 and 14 seconds?	38. What is the probability that a runner finished in under 12.7 seconds?
39. What is the probability that it took a runner more than 13.5 seconds?	40. Lana finished the race in 13.2 seconds. If 84 other girls ran in the event, approximately how many runners did she beat?

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
FUNDAMENTAL COUNTING PRINCIPLE	If decision M can be made <u>m</u> ways and decision N can be made <u>n</u> ways, then the two decisions can be made <u>$m \cdot n$</u> ways.
EXAMPLES	1. The ice cream shop offers a choice of a 3 cone sizes, 15 flavors, and 8 toppings. How many cones are possible if you can only choose one flavor and one topping? $3 \cdot 15 \cdot 8 = \boxed{360}$ 2. Virginia license plates consist of three letters followed by 4 digits. How many different license plates are possible? $26 \cdot 26 \cdot 26 \cdot 10 \cdot 10 \cdot 10 \cdot 10 = 26^3 \cdot 10^4 = \boxed{175,760,000}$ 3. Nick's science test has a section with 12 true or false questions. How many different ways can Nick answer these questions? $2^{12} = \boxed{4096}$ 4. To enter their home, the Clayton family enters a 4-digit code. How many codes are possible? $10^4 = \boxed{10,000}$
FACTORIALS $n!$	The <u>product</u> of all natural numbers from <u>n</u> to <u>1</u>. Examples: $9! = 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 362,880$ $12! = 479,001,600$ $8! = 40,320$ $\frac{17!}{11!} = 8,910,720$ $*0! = 1$
PERMUTATIONS ${}_n P_r$	An arrangement or line up of objects in which <u>order</u> <u>matters</u>! Permutation Formula: ${}_n P_r = \frac{n!}{(n-r)!}$ n = total number of objects available r = number of objects to use for the arrangement *Important Shortcut: ${}_n P_n = n!$
EXAMPLES	5. How many ways can you arrange the letters in the word DINOSAUR? ${}_8 P_8 = 8! = \boxed{40,320}$ 6. Seven students are competing in a geography bee. How many ways can they finish in first, second, and third place? ${}_7 P_3 = \frac{7!}{(7-3)!} = \frac{7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{4 \cdot 3 \cdot 2 \cdot 1} = \boxed{210}$

	7. Melanie is taking four classes this semester: American History, Algebra 2, AP English, and Chemistry. How many ways can these four classes be arranged on her schedule? ${}_4P_4 = 4! = \boxed{24}$ 8. There are 16 players on the baseball team. How many ways can the coach make a 9-player batting order? ${}_{16}P_9 = \frac{16!}{(16-9)!} = \boxed{4,151,347,200}$ 9. There are 28 students in Mr. Miller's homeroom. How many ways can the students elect a student council representative and alternate? ${}_{28}P_2 = \frac{28!}{(28-2)!} = \boxed{756}$
PERMUTATIONS <i>with Repetition</i>	10. How many different 10-letter arrangements are possible using the letters in the word <u>AUTOMOBILE</u>? $\frac{10!}{2!} = \boxed{1,814,400}$ 11. How many different 9-letter arrangements are possible using the letters in the word <u>DISAPPEAR</u>? $\frac{9!}{2! \cdot 2!} = \boxed{90,720}$
COMBINATIONS ${}_nC_r$	A group of objects in which <u>order</u> <u>doesn't</u> <u>matter</u> ! Combination Formula: ${}_nC_r = \frac{n!}{r! (n-r)!}$ n = total number of objects available r = number of objects to use for the arrangement *Important Shortcut: ${}_nC_n = \boxed{1}$
EXAMPLES	12. Natalie has 16 close friends. How many ways can she choose 5 to be bridesmaids in her wedding? ${}_{16}C_5 = \frac{16!}{5! (16-5)!} = \boxed{4,368}$ 13. There are 24 students in Kyle's kindergarten class. How many ways can he choose eight to attend his birthday party? ${}_{24}C_8 = \frac{24!}{8! (24-8)!} = \boxed{735,471}$ 14. Abby is adopting kittens from the pet store. If there are 18 kittens, how many ways can she choose two? ${}_{18}C_2 = \frac{18!}{2! (18-2)!} = \boxed{153}$ 15. There are twelve employees at the sub shop. How many ways can the manager choose four for the Sunday evening shift? ${}_{12}C_4 = \frac{12!}{4! (12-4)!} = \boxed{495}$ 16. There are 85 players on the football team. How many ways can the coach choose three to represent the team in the coin toss? ${}_{85}C_3 = \frac{85!}{3! (85-3)!} = \boxed{98,770}$

Permutation or Combination?

Directions: Recall that a **permutation** is an **arrangement** with a specific order, while a **combination** is a **group** with no specific order. Determine whether the example represents a permutation or combination, then solve.

Example	P or C?	Answer
1 How many ways can 4 candy bars be chosen from a store that sells 30 candy bars?	C	${}_{30}C_4 = \frac{30!}{4! \cdot 26!} = \boxed{27,405}$
2 How many ways can 13 students line up for lunch?	P	${}_{13}P_{13} = 13! = \boxed{6,227,020,800}$
3 How many ways can you make a 3-letter arrangements out of the letters in the word TRAPEZOID .	P	${}_9P_3 = \frac{9!}{6!} = \boxed{504}$
4 How many ways can you choose 2 books from a shelf of 40 books?	C	${}_{40}C_2 = \frac{40!}{2! \cdot 38!} = \boxed{780}$
5 How many ways can 12 swimmers finish in first, second, and third place?	P	${}_{12}P_3 = \frac{12!}{9!} = \boxed{1320}$
6 How many ways can Mrs. Sullivan choose two students from 27 to help put away calculators at the end of class?	C	${}_{27}C_2 = \frac{27!}{2! \cdot 25!} = \boxed{351}$
7 You have enough tickets to play 6 different games at the amusement park. If there are 14 games, how many ways can you choose six?	C	${}_{14}C_6 = \frac{14!}{6! \cdot 8!} = \boxed{3003}$
8 How many different ways can 9 trumpet players in the marching band line up?	P	${}_9P_9 = 9! = \boxed{362,880}$
9 Seven students worked together on a project. How many ways can their teacher choose four to present the project?	C	${}_7C_4 = \frac{7!}{4! \cdot 3!} = \boxed{35}$
10 There are 18 offensive players on the hockey team. How many ways can the coach choose a left wing, center, and right wing to start the game?	P	${}_{18}P_3 = \frac{18!}{15!} = \boxed{4896}$
11 How many different 12-letter arrangements can be made using the letters in the word INDIANAPOLIS ?	P	$\frac{12!}{2! \cdot 2! \cdot 3!} = \boxed{199,584,000}$
12 There are 26 gold fish in the tank at the store. How many ways can Ben choose five?	C	${}_{26}C_5 = \frac{26!}{5! \cdot 21!} = \boxed{65,780}$

Name: _____

Unit 11: Probability & Statistics

Date: _____ Bell: _____

Homework 1: Fundamental Counting Principle,
Permutations & Combinations**** This is a 2-page document! ******Directions:** Find the total number of outcomes that are possible.

1. In a school building, there are 8 exterior doors and 12 stairways to enter the second floor. Find the total number of ways to reach the second floor.

$$8 \cdot 12 = \boxed{96}$$

2. Kendall went shopping and purchased 7 shirts, 3 pairs of pants, 2 jackets. How many different outfits are possible?

$$7 \cdot 3 \cdot 2 = \boxed{42}$$

3. An internet passcode consists of a digit followed by a letter, followed by another digit. Assuming the digits are 0-9. How many different passcodes are possible?

$$10 \cdot 26 \cdot 10 = \boxed{2,600}$$

4. Grant is rolling a standard six-sided number die eight times. How many outcomes are possible?

$$6^8 = \boxed{1,679,616}$$

5. A quiz has eight multiple choice questions with four options for each (A, B, C, and D). How many ways are there to answer the questions?

$$4^8 = \boxed{65,536}$$

6. When Jack bought his new truck, there were 96 different ways his truck could be equipped. He had four choices of engines and two choices of transmissions. If the only other choice was color, how colors were available?

$$4 \cdot 2 \cdot x = 96$$

$$8x = 96$$

$$\boxed{x = 12}$$

Directions: Evaluate the following.7. $6!$

$$720$$

8. $\frac{16!}{7!}$

$$4,151,347,200$$

9. $\frac{7!}{10!}$

$$0.0013\overline{8}$$

10. $\frac{2! \cdot 9!}{(10-7)!}$

$$\frac{725760}{6} = 120,960$$

11. ${}_{20}P_4$

$$116,280$$

12. ${}_{15}C_8$

$$6,435$$

13. ${}_{11}P_7$

$$98,017,920$$

14. ${}_{27}C_{22}$

$$80,730$$

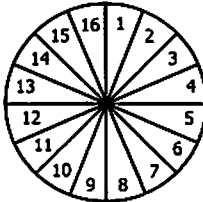
15. Find ${}_9P_9$ and ${}_9C_9$. Why do the answers differ? Explain.

$${}_9P_9 = 362,880 \quad {}_9C_9 = 1$$

The combination is significantly lower because the order does not matter.

Directions: Determine whether the situations represent permutations or combinations. Then solve.		
Example	P/C	Solve
16. Scott has 7 chores to complete this Saturday. How many ways can he arrange the order in which he does them?	P	$7P_7 = 7!$ $= \boxed{5040}$
17. There are 16 different colored markers in the bucket. How many ways can Kelly choose five of them?	C	$16C_5 = \frac{16!}{5! 11!}$ $= \boxed{4368}$
18. How many ways are there to elect a President, Vice President, Secretary, and Treasurer, from a club with 32 members?	P	$32P_4 = \frac{32!}{28!}$ $= \boxed{863,040}$
19. The dance company is choosing 3 new dancers from a group of 25 who try out. How many ways can they choose the new dancers?	C	$25C_3 = \frac{25!}{3! 22!}$ $= \boxed{2300}$
20. There are ten toppings available to make an ice cream sundae. How many ways can Max choose two?	C	$10C_2 = \frac{10!}{2! 8!}$ $= \boxed{45}$
21. How many different 11-letter arrangements can be made using the letters in the word FIRECRACKER ? ·≡≡≡≡≡≡≡≡≡	P	$\frac{11!}{2! \cdot 2! \cdot 3!} = \boxed{1,663,200}$
22. Mindy is purchasing songs from an album on iTunes. If the album has 14 songs, how many ways can she choose 6?	C	$14C_6 = \frac{14!}{6! 8!}$ $= \boxed{3003}$
23. Carl is choosing a four-character password for his cell phone. The password can contain letters and digits only, and can't include repeating characters. How many passwords are possible?	P	$36P_4 = \frac{36!}{32!}$ $= \boxed{1,413,720}$
24. Mr. Pratt has 26 students in his math class. He has three prizes to give away: a pencil, an eraser, and a no homework coupon. How many ways can he choose three students to win these awards?	P	$26P_3 = \frac{26!}{23!}$ $= \boxed{15,600}$
25. How many ways are there to choose seven cards from a deck of 52?	C	$52C_7 = \frac{52!}{7! 45!}$ $= \boxed{133,784,560}$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
Theoretical Probability	- Probability is the measure of how <u>likely</u> an event is to occur. - The set of all possible outcomes is called the sample space. - For equally likely outcomes, the theoretical probability of an event, $P(E)$, is the <u>ratio</u> of the number of favorable outcomes to the total number of outcomes possible.	
Simple Events The probability of one event.	1. A jar contains 32 red marbles and 28 blue marbles. What is the probability that a randomly selected marble is red? $\frac{32}{60} = \boxed{\frac{8}{15}}$	2. A letter in the word RESTORATION is randomly selected. What is the probability of selecting a vowel? $\boxed{\frac{5}{11}}$
	3. A day in the month of January is randomly selected. What is the probability of selecting a prime number? $\boxed{\frac{11}{31}}$	4. Two dice are rolled. What is the probability that the sum of the two dice on the next roll is at least 9? $\frac{10}{36} = \boxed{\frac{5}{18}}$
	5. What is the probability of drawing a heart or a club from a standard deck of cards? $\frac{26}{52} = \boxed{\frac{1}{2}}$	6. There are 8 books lettered A-H on the shelf. If Scott randomly chose two books, what is the probability that he chose books A and B? $\frac{1}{{}^8C_2} = \boxed{\frac{1}{28}}$
Complement of an Event 	The complement of an event is the probability of the event <u>not</u> happening. Since the sum of all probabilities in sample space is <u>1</u> , the probability of an event not happening is $P(\sim E) = \underline{1 - P(E)}$.	
	7. The probability that it will snow tomorrow is $\frac{7}{20}$. What is the probability that it will not snow? $1 - \frac{7}{20} = \boxed{\frac{13}{20}}$	8. A month of the year is randomly selected. What is the probability of getting a month that does not begin with the letter A? $1 - \frac{2}{12} = \boxed{\frac{5}{6}}$
	9. If the spinner to the left is spun, find the probability that it lands on a number that is not prime. $1 - \frac{6}{16} = \boxed{\frac{5}{8}}$	10. Two dice are rolled. What is the probability of not getting doubles? $1 - \frac{6}{36} = \boxed{\frac{5}{6}}$

Compound Events The probability of two or more simple events.	Independent Events When the outcome of one event <u>does not</u> affect the outcome of the other event. $P(A \text{ and } B) =$ $P(A) \cdot P(B)$	Dependent Events When the outcome of one event <u>does</u> affect the outcome of the other event. $P(A \text{ and } B) =$ $P(A) \cdot P(B, \text{after } A \text{ occurred})$
Independent Events	11. A die is rolled 3 times. What is the probability of getting 1's on each roll? $\frac{1}{6} \cdot \frac{1}{6} \cdot \frac{1}{6} = \boxed{\frac{1}{216}}$	12. A coin is tossed, then a day of the week is selected. What is the probability of getting tails then a day starting with the letter T? $\frac{1}{2} \cdot \frac{2}{7} = \boxed{\frac{1}{7}}$
	A bag contains 8 red crayons, 14 purple crayons, 6 yellow crayons, and 4 green crayons. A crayon is selected, replaced, then another is selected. Find each probability.	
	13. $P(\text{purple then yellow})$ $\frac{14}{32} \cdot \frac{6}{32} = \boxed{\frac{21}{256}}$	14. $P(\text{green then red})$ $\frac{4}{32} \cdot \frac{8}{32} = \boxed{\frac{1}{32}}$
	15. $P(\text{two purples})$ $\frac{14}{32} \cdot \frac{14}{32} = \boxed{\frac{49}{256}}$	16. $P(\text{two yellows})$ $\frac{6}{32} \cdot \frac{6}{32} = \boxed{\frac{9}{256}}$
Dependent Events	Using the same example from above, assume once a crayon is selected, it is NOT replaced. Find each probability.	
	17. $P(\text{yellow then red})$ $\frac{6}{32} \cdot \frac{8}{31} = \boxed{\frac{3}{62}}$	18. $P(\text{purple then green})$ $\frac{14}{32} \cdot \frac{4}{31} = \boxed{\frac{7}{124}}$
	19. $P(\text{two reds})$ $\frac{8}{32} \cdot \frac{7}{31} = \boxed{\frac{7}{124}}$	20. $P(\text{two greens})$ $\frac{4}{32} \cdot \frac{3}{31} = \boxed{\frac{3}{248}}$
	21. A card is drawn from a standard deck, not replaced, and another is drawn. What is the probability of choosing a heart then a spade? $\frac{13}{52} \cdot \frac{13}{51} = \boxed{\frac{13}{204}}$	22. Jack had four Snicker bars and 8 Mars bars. He randomly chose a piece of candy, ate it, then chose another. What is the probability that both candy bars were Snickers? $\frac{4}{12} \cdot \frac{3}{11} = \boxed{\frac{1}{11}}$

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Homework 2: Theoretical
Probability**** This is a 2-page document! ******Part I: Simple Probability****Use for questions 1-3: A random two-digit number (10-99) is drawn.****Find each probability.**1. $P(32)$

$$\frac{1}{90}$$

2. $P(\text{odd number})$

$$\frac{45}{90} = \frac{1}{2}$$

3. $P(\text{a multiple of 5})$

$$\frac{18}{90} = \frac{1}{5}$$

Use for questions 4-6: A letter is randomly chosen from the word CANDLESTICK.**Find each probability.**4. $P(\text{a vowel})$

$$\frac{3}{11}$$

5. $P(N \text{ or } S)$

$$\frac{2}{11}$$

6. $P(\text{not } C)$

$$1 - \frac{2}{11} = \frac{9}{11}$$

7. Three coins are tossed. Find the probability that two land on heads.

H H H T H H
 H H T T T H
 H T H T H T
 H T T T T T

$$\frac{3}{8}$$

8. A month is randomly chosen. What is the probability that the month chosen has less than 31 days?

$$1 - \frac{7}{12} = \frac{5}{12}$$

9. What is the probability of drawing a 9 or diamond from a standard deck of cards?

$$\frac{16}{52} = \frac{4}{13}$$

10. Credit cards place a three-digit security code on the back of cards. What is the probability that a code starts with the number 7?

$$\frac{1 \cdot 10 \cdot 10}{10 \cdot 10 \cdot 10} = \frac{1}{10}$$

11. Two dice are rolled. What is the probability of not getting doubles?

$$1 - \frac{6}{36} = \frac{5}{6}$$

12. Mikayla has the following songs on her iPod: 14 Taylor Swift songs, 16 Meghan Trainor songs, and 17 Katy Perry songs. What is the probability that the next song that plays is not Katy Perry?

$$1 - \frac{17}{47} = \frac{30}{47}$$

Part II: Compound Probability

13. A dice is rolled, then a coin is tossed. What is the probability of getting a 5 then tails?

$$\frac{1}{6} \cdot \frac{1}{2} = \frac{1}{12}$$

14. A coin is tossed, then a number 1-10 is chosen at random. What is the probability of getting heads then a number less than 4?

$$\frac{1}{2} \cdot \frac{3}{10} = \frac{3}{20}$$

15. Natalie guessed on the last four true or false questions on her math quiz. What is the probability that she got all four questions correct?

$$\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \left(\frac{1}{2}\right)^4 = \boxed{\frac{1}{16}}$$

16. A card is drawn from a standard deck and a letter is chosen from the word **INCREDIBLE**. What is the probability of drawing a king then getting an I?

$$\frac{4}{52} \cdot \frac{2}{10} = \boxed{\frac{1}{65}}$$

Use for questions 17-20: A bag contains 30 lottery balls numbered 1-30. A ball is selected, replaced, then another is drawn. Find each probability.

17. $P(\text{and even, then odd})$

$$\frac{15}{30} \cdot \frac{15}{30} = \boxed{\frac{1}{4}}$$

18. $P(7, \text{ then a number greater than } 16)$

$$\frac{1}{30} \cdot \frac{14}{30} = \boxed{\frac{7}{450}}$$

19. $P(\text{a multiple of } 5, \text{ then a prime number})$

$$\frac{6}{30} \cdot \frac{10}{30} = \boxed{\frac{1}{15}}$$

20. $P(\text{two even numbers})$

$$\frac{15}{30} \cdot \frac{15}{30} = \boxed{\frac{1}{4}}$$

Use for questions 21-24: A bag contains 30 lottery balls numbered 1-30. A ball is selected, NOT replaced, then another is drawn. Find each probability.

21. $P(\text{a 2-digit number, then } 4)$

$$\frac{21}{30} \cdot \frac{1}{29} = \boxed{\frac{7}{290}}$$

22. $P(19, \text{ then a multiple of } 4)$

$$\frac{1}{30} \cdot \frac{7}{29} = \boxed{\frac{7}{870}}$$

23. $P(24, \text{ then a number less than } 15)$

$$\frac{1}{30} \cdot \frac{14}{29} = \boxed{\frac{7}{435}}$$

24. $P(\text{two perfect squares})$

$$\frac{5}{30} \cdot \frac{4}{29} = \boxed{\frac{2}{87}}$$

25. A football team has 5 freshman, 8 sophomores, 11 juniors, and 16 seniors. If two are chosen at random to participate in the coin toss, what the probability that both players chosen will be seniors?

$$\frac{16}{40} \cdot \frac{15}{39} = \boxed{\frac{2}{13}}$$

26. Ryan's mom randomly chooses two days each week for Ryan to do his chores. What is the probability that she picks Saturday and Sunday?

$$\frac{2}{7} \cdot \frac{1}{6} = \boxed{\frac{1}{21}}$$

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Main Ideas/Questions	Notes/Examples									
Conditional Probability	- Given events A and B, conditional probability is the probability of event B occurring, knowing that event A has <u>already</u> <u>occurred</u>. - Notation: <u>$P(B/A)$</u> - Read as: <u>the probability of B given A</u> Conditional Probability Formula: $P(B A) = \frac{P(A \text{ and } B)}{P(A)}$ 									
Examples	1. Mitchell drew a card from a standard deck of playing cards. What is the probability that he drew a queen, given that the card was red? $\frac{\frac{26}{52} \cdot \frac{4}{52}}{\frac{26}{52}} = \boxed{\frac{1}{13}}$ 2. A number from 1-100 is randomly selected. What is the probability that it is a perfect square, given that it is an odd number? $\frac{\frac{50}{100} \cdot \frac{10}{100}}{\frac{50}{100}} = \boxed{\frac{1}{10}}$ 3. There are 62 people that take yoga class and 48 people that take spinning class at the gym. Fifteen people that take yoga also take spinning. If a person from these groups is selected at random, find the probability that he or she is takes yoga class, if you know they take spinning class. $\frac{15}{48} = \boxed{\frac{5}{16}}$ yoga 4. Of the 30 students in Mrs. Smith's math class, 13 are athletes and 9 are in band. Three of the athletes are also in band. If one student is chosen at random, find the probability that this student is in band, if it is known that they are <u>not</u> an athlete. $\boxed{\frac{6}{17}}$ Athlete Band									
Two-Way Tables	A table that records data that pertains to two different categories. Example: The table shows the results of a poll of 50 randomly selected students who were asked whether the prefer to watch hockey or football. Find each probability. <table border="1" style="border-collapse: collapse; text-align: center; margin-right: 20px;"> <tr> <th style="padding: 5px;"></th><th style="padding: 5px;">Boys</th><th style="padding: 5px;">Girls</th></tr> <tr> <td style="padding: 5px;">Prefer Football</td><td style="padding: 5px;">18</td><td style="padding: 5px;">6</td></tr> <tr> <td style="padding: 5px;">Prefer Hockey</td><td style="padding: 5px;">10</td><td style="padding: 5px;">16</td></tr> </table> a) $P(\text{prefer football} \text{girl})$ $\frac{6}{22} = \boxed{\frac{3}{11}}$ b) $P(\text{boy} \text{hockey})$ $\frac{10}{26} = \boxed{\frac{5}{13}}$		Boys	Girls	Prefer Football	18	6	Prefer Hockey	10	16
	Boys	Girls								
Prefer Football	18	6								
Prefer Hockey	10	16								

Directions: Find each probability using the table.

5. The table shows the results of a poll of 150 randomly selected middle school students who were asked if they take French or Spanish.

	French	Spanish
6 th Grade	19	23
7 th Grade	25	12
8 th Grade	38	33

a) $P(\text{a 7th grader who takes French})$

$$\frac{25}{150} = \frac{1}{6}$$

b) $P(\text{8th grader} \mid \text{takes Spanish})$

$$\frac{33}{68}$$

c) $P(\text{takes French} \mid \text{6th grader})$

$$\frac{19}{42}$$

d) $P(\text{7th grader} \mid \text{takes French})$

$$\frac{25}{82}$$

6. The table shows the results of a poll of randomly selected juniors and seniors who were asked if they attended prom.

		Juniors	Seniors
Prom	Yes	28	97
	No	56	19

(200 total)

a) $P(\text{a junior that did not attend prom})$

$$\frac{56}{200} = \frac{7}{25}$$

b) $P(\text{did not attend prom} \mid \text{senior})$

$$\frac{19}{116}$$

c) $P(\text{junior} \mid \text{attended prom})$

$$\frac{28}{125}$$

Frequencies

- **Joint Relative Frequencies:** The values in each category divided by total number of values.
- **Marginal Relative Frequencies:** Found by totaling each row and column.

Examples

7. The table below shows the number of first class and coach class passengers on a plane who checked one bag, two bags, or no bags. Find the joint relative and marginal relative frequencies, then find each probability.

	First Class	Coach Class	Total
2 bags	16	52	68
1 bag	12	60	72
0 bags	2	18	20
Total	30	130	160

(160 total)

a) $P(\text{a coach class passenger that did not check any bags})$

$$0.1125 = 11.25\%$$

b) $P(\text{a first class passenger that checked at least one bag})$

$$0.1 + 0.075 = 0.175 \quad 17.5\%$$

c) $P(\text{coach class} \mid \text{checked one bag})$

$$\frac{0.375}{0.45} = 0.8\bar{3} \quad 83\bar{3}\%$$

d) $P(\text{checked two bags} \mid \text{first class})$

$$\frac{0.1}{0.1875} = 0.5\bar{3} \quad 53\bar{3}\%$$

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Homework 3: Conditional Probability

**** This is a 2-page document! ******Use for questions 1-2:** A bucket contains 50 lottery balls numbered 1-50. One is drawn at random. Find each probability.1. $P(\text{multiple of 6} \mid \text{2-digit number})$

$$\frac{7}{41}$$

2. $P(\text{at least 20} \mid \text{prime number})$

$$\frac{7}{15}$$

3. Marti rolls two dice. What is the probability that the sum of the dice is 7, given that the first die is showing a 2?

$$\frac{1}{6}$$

4. Blake randomly chose a letter from alphabet. What is the probability that this letter has at least one line of symmetry, given that it is a consonant?

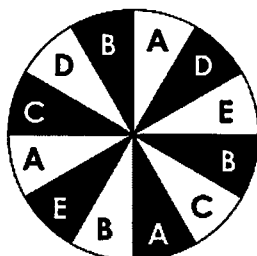
$$\frac{11}{21}$$

5. A card is randomly selected from a standard deck of playing cards. Find the probability that it is a face card, given that a black card is drawn.

$$\frac{6}{26} = \frac{3}{13}$$

6. A month of the year is randomly chosen. Find the probability that it has no more than 30 days, given that it starts with the letter A.

$$\frac{1}{2}$$

Use for 7-9: The wheel below is spun. Find each probability.7. $P(\text{black} \mid \text{A})$

$$\frac{1}{3}$$

8. $P(\text{C} \mid \text{white})$

$$\frac{1}{6}$$

9. $P(\text{black} \mid \text{B or E})$

$$\frac{3}{5}$$

10. Out of the 125 children at summer camp, 45 signed up for swimming and 38 signed up for arts and crafts. Twelve students who signed up for swimming also signed up for arts and crafts. If a child is randomly selected, what is the probability that they are signed up for swimming, if it is known that they did not sign up for arts and crafts?

$$\frac{33}{87}$$

11. Out of the 56 players on the football team, 24 are on honor roll and 18 have perfect attendance. Seven who are on honor roll also have perfect attendance. If a player is chosen at random, what is the probability that they are on honor roll, if it is known that they also have perfect attendance?

$$\frac{7}{18}$$

Directions: Find each probability using the table.

- 12.** The table below shows the number of students that do or do not have their own car and whether they have part time jobs.

		Part Time Job	
		Yes	No
Car	Yes	78	18
	No	30	24

- a) $P(\text{a student with a part-time job without a car})$

$$\frac{30}{150} = \frac{1}{5}$$

- b) $P(\text{no car} | \text{does not have a part-time job})$

$$\frac{24}{42} = \frac{4}{7}$$

- c) $P(\text{part-time job} | \text{car})$

$$\frac{78}{96} = \frac{13}{16}$$

- 13.** The table below shows the fate of the first, second, and third class passengers on the Titanic.

	Survived	Died
1 st Class	199	120
2 nd Class	117	155
3 rd Class	172	537

- a) $P(\text{a 3rd class survivor})$

$$\frac{172}{1300} = \frac{43}{325}$$

- b) $P(1^{\text{st}} \text{ class} | \text{died})$

$$\frac{120}{812} = \frac{30}{203}$$

- c) $P(\text{survived} | 2^{\text{nd}} \text{ class})$

$$\frac{117}{272}$$

- d) $P(1^{\text{st}} \text{ or } 3^{\text{rd}} \text{ class} | \text{survived})$

$$\frac{371}{488}$$

- 14.** The table below represents the GPA of a group of undergrad students and whether or not they will be attending grad school. Find the joint relative and marginal relative frequencies.

		Attending Grad School	
		Yes	No
GPA	≥ 3.0	96	18
	< 3.0	32	54

		Attending Grad School		
		Yes	No	Total
GPA	≥ 3.0	0.48	0.09	0.57
	< 3.0	0.16	0.27	0.43
Total		0.64	0.36	1

If a student is chosen at random, find each probability:

- a) $P(\text{not attending grad school})$

$$0.36 \quad 36\%$$

- b) $P(< 3.0 \text{ GPA and not attending grad school})$

$$0.27 \quad 27\%$$

- c) $P(\text{not attending grad school} | \geq 3.0 \text{ GPA})$

$$\frac{0.09}{0.57} \approx .1597 \quad 15.97\%$$

- d) $P(< 3.0 \text{ GPA} | \text{attending grad school})$

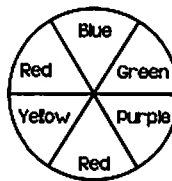
$$\frac{.16}{.64} = 0.25 \quad 25\%$$

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Main Ideas/Questions	Notes/Examples
PASCAL'S TRIANGLE	Pascal's triangle was a pattern of numbers that was discovered in the 13th century. Each number in Pascal's triangle is the sum of the two numbers diagonally above it. All outside numbers are 1. Complete rows 4 and 5 of Pascal's triangle below: Row 0 → 1 Row 1 → 1 1 Row 2 → 1 2 1 Row 3 → 1 3 3 1 Row 4 → 1 4 6 4 1 Row 5 → 1 5 10 10 5 1
<i>Expanding Binomials</i>	Expand the binomial $(a + b)^3$: $(a+b)(a+b)(a+b)$ $(a+b)(a^2 + 2ab + b^2)$ $a^3 + 2a^2b + ab^2 + a^2b + 2ab^2 + b^3$ What do you notice about the coefficients? $= \underline{a^3} + \underline{3a^2b} + \underline{3ab^2} + \underline{b^3}$ They are the coefficients from row 3!
THE BINOMIAL THEOREM	If n is a natural number, then $(a + b)^n =$ ${}_nC_0 \cdot a^n b^0 + {}_nC_1 \cdot a^{n-1} b^1 + {}_nC_2 \cdot a^{n-2} b^2 + \dots + {}_nC_n \cdot a^0 b^n = \sum_{k=0}^n {}_nC_k \cdot a^{n-k} b^k$
<i>Examples</i>	Directions: Use the binomial theorem to expand each binomial. 1. $(a+b)^5$ ${}_5C_0 a^5 b^0 + {}_5C_1 a^4 b^1 + {}_5C_2 a^3 b^2 + {}_5C_3 a^2 b^3 + {}_5C_4 a^1 b^4 + {}_5C_5 a^0 b^5$ $= a^5 + 5a^4b + 10a^3b^2 + 10a^2b^3 + 5ab^4 + b^5$ 2. $(x+y)^7$ ${}_7C_0 x^7 y^0 + {}_7C_1 x^6 y^1 + {}_7C_2 x^5 y^2 + {}_7C_3 x^4 y^3 + {}_7C_4 x^3 y^4 + {}_7C_5 x^2 y^5 + {}_7C_6 x^1 y^6 + {}_7C_7 x^0 y^7$ $= x^7 + 7x^6y + 21x^5y^2 + 35x^4y^3 + 35x^3y^4 + 21x^2y^5 + 7xy^6 + y^7$

	3. $(c+d)^{10}$ ${}_{10}C_0 c^{10}d^0 + {}_{10}C_1 c^9d^1 + {}_{10}C_2 c^8d^2 + {}_{10}C_3 c^7d^3 + {}_{10}C_4 c^6d^4 + {}_{10}C_5 c^5d^5 + {}_{10}C_6 c^4d^6 + {}_{10}C_7 c^3d^7 + {}_{10}C_8 c^2d^8 + {}_{10}C_9 c^1d^9 + {}_{10}C_{10} c^0d^{10}$ $= c^{10} + 10c^9d + 45c^8d^2 + 120c^7d^3 + 210c^6d^4 + 252c^5d^5 + 210c^4d^6 + 120c^3d^7 + 45c^2d^8 + 10cd^9 + d^{10}$
Coefficients Other than 1	4. $(x-3)^6$ ${}_6C_0 x^6(-3)^0 + {}_6C_1 x^5(-3)^1 + {}_6C_2 x^4(-3)^2 + {}_6C_3 x^3(-3)^3 + {}_6C_4 x^2(-3)^4 + {}_6C_5 x^1(-3)^5 + {}_6C_6 x^0(-3)^6$ $= x^6 - 18x^5 + 135x^4 - 540x^3 + 1215x^2 - 1458x + 729$
	5. $(2m+n)^7$ ${}_7C_0 (2m)^7n^0 + {}_7C_1 (2m)^6n^1 + {}_7C_2 (2m)^5n^2 + {}_7C_3 (2m)^4n^3 + {}_7C_4 (2m)^3n^4 + {}_7C_5 (2m)^2n^5 + {}_7C_6 (2m)^1n^6 + {}_7C_7 (2m)^0n^7$ $= 128m^7 + 448m^6n + 672m^5n^2 + 560m^4n^3 + 280m^3n^4 + 84m^2n^5 + 14mn^6 + n^7$
	6. $(k+2)^8$ ${}_8C_0 k^8(2)^0 + {}_8C_1 k^7(2)^1 + {}_8C_2 k^6(2)^2 + {}_8C_3 k^5(2)^3 + {}_8C_4 k^4(2)^4 + {}_8C_5 k^3(2)^5 + {}_8C_6 k^2(2)^6 + {}_8C_7 k^1(2)^7 + {}_8C_8 k^0(2)^8$ $= k^8 + 16k^7 + 112k^6 + 448k^5 + 1120k^4 + 1792k^3 + 1792k^2 + 1024k + 256$
	7. $(3p-2q)^5$ ${}_5C_0 (3p)^5(-2q)^0 + {}_5C_1 (3p)^4(-2q)^1 + {}_5C_2 (3p)^3(-2q)^2 + {}_5C_3 (3p)^2(-2q)^3 + {}_5C_4 (3p)^1(-2q)^4 + {}_5C_5 (3p)^0(-2q)^5$ $= 243p^5 - 810p^4q + 1080p^3q^2 - 720p^2q^3 + 240pq^4 - 32q^5$
Observations	In the binomial expansion of $(a+b)^n$: - The total number of terms is always <u>$n+1$</u>. - The exponent of a in the first term is <u>n</u>. - The exponent of b in the last term is <u>n</u>. - The exponent of a <u>decreases by 1</u> from left to right. - The exponent of b <u>increases by 1</u> from left to right. - The sum of the exponents in each term is <u>n</u>. - The coefficients are <u>symmetrical</u> and follow the <u>n^{th} row of Pascal's Triangle</u>.

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Main Ideas/Questions	Notes/Examples															
Binomial Experiments	A binomial experiment consists of n independent trials in which: - There are only two outcomes: success and failure - The probability of success p is the same in every trial. - The probability of failure q is the same in every trial. - The probability of success is p. - The probability of failure is q (or $1-p$)															
Examples	<table><thead><tr><th>Experiment</th><th>Success</th><th>Failure</th><th>p</th><th>q</th></tr></thead><tbody><tr><td>Randomly selecting a month 30 times</td><td>January</td><td>Not January</td><td>$\frac{1}{12}$</td><td>$1-\frac{1}{12}=\frac{11}{12}$</td></tr><tr><td>Randomly selecting a card 100 times</td><td>Heart</td><td>Not a Heart</td><td>$\frac{1}{4}$</td><td>$1-\frac{1}{4}=\frac{3}{4}$</td></tr></tbody></table>	Experiment	Success	Failure	p	q	Randomly selecting a month 30 times	January	Not January	$\frac{1}{12}$	$1-\frac{1}{12}=\frac{11}{12}$	Randomly selecting a card 100 times	Heart	Not a Heart	$\frac{1}{4}$	$1-\frac{1}{4}=\frac{3}{4}$
Experiment	Success	Failure	p	q												
Randomly selecting a month 30 times	January	Not January	$\frac{1}{12}$	$1-\frac{1}{12}=\frac{11}{12}$												
Randomly selecting a card 100 times	Heart	Not a Heart	$\frac{1}{4}$	$1-\frac{1}{4}=\frac{3}{4}$												
Binomial Probability	If a binomial experiment has n trials in which p is the probability of success and q is the probability of failure, then the binomial probability that there will be exactly r successes is: ${}_nC_r \cdot p^r \cdot q^{n-r}$															
Examples (with an exact number of successes)	1. Mark tosses a coin 25 times. What is the probability that it lands on heads exactly seven times? $n=25, p=\frac{1}{2}, q=\frac{1}{2}, r=7$ ${}_{25}C_7 \left(\frac{1}{2}\right)^7 \cdot \left(\frac{1}{2}\right)^{18} = .0143$ 1.43% 2. The spinner below is spun 10 times. What is the probability that it lands on red exactly six times?  $n=10, p=\frac{1}{3}, q=\frac{2}{3}, r=6$ ${}_{10}C_6 \left(\frac{1}{3}\right)^6 \cdot \left(\frac{2}{3}\right)^4 = .0569$ 5.69% 3. Kate took a multiple choice test and guessed on the last 5 questions. What is the probability that she got exactly three questions correct? $n=5, p=\frac{1}{4}, q=\frac{3}{4}, r=3$ ${}_5C_3 \left(\frac{1}{4}\right)^3 \cdot \left(\frac{3}{4}\right)^2 = .0879$ 8.79% 4. Chad randomly chooses a card from a standard deck 30 times. What is the probability that he gets an diamond exactly four times? $n=30, p=\frac{1}{4}, q=\frac{3}{4}, r=4$ ${}_{30}C_4 \left(\frac{1}{4}\right)^4 \cdot \left(\frac{3}{4}\right)^{26} = .0604$ 6.04%															

	5. There are 30 balls numbered 1-30 in a bag. One is randomly selected 12 times. What is the probability of getting a prime number exactly five times? $n=12, p=\frac{10}{30}, q=\frac{20}{30}, r=5$ ${}_{12}C_5 \left(\frac{1}{3}\right)^5 \cdot \left(\frac{2}{3}\right)^7 = .1908$ 19.08%	6. The probability of customers at a store receiving a coupon printed on their receipt is 1 in 8. If 20 customers visit the store today, what is the probability that exactly three of them receive coupons? $n=20, p=\frac{1}{8}, q=\frac{7}{8}, r=3$ ${}_{20}C_3 \left(\frac{1}{8}\right)^3 \cdot \left(\frac{7}{8}\right)^{17} = .2300$ 23%
	7. In a game of chance, the probability of winning is 30%. If 15 games are played, find the probability of winning exactly six games. $n=15, p=0.3, q=0.7, r=6$ ${}_{15}C_6 (0.3)^6 \cdot (0.7)^9 = .1472$ 14.72%	8. A certain traffic light is red for 20 seconds, yellow for 4 seconds, and green for 36 seconds. Find the probability that of the next 12 cars that randomly arrive at the light, exactly 4 will be stopped by a red light. $n=12, p=\frac{20}{60}, q=\frac{40}{60}, r=4$ ${}_{12}C_4 \left(\frac{1}{3}\right)^4 \cdot \left(\frac{2}{3}\right)^8 = .2384$ 23.84%
Examples (with at least, at most, etc.)	9. Kyle tosses a coin 10 times. What is the probability that he gets tails at least 8 times? ${}_{10}C_8 \left(\frac{1}{2}\right)^8 \left(\frac{1}{2}\right)^2 = 0.0439$ ${}_{10}C_9 \left(\frac{1}{2}\right)^9 \left(\frac{1}{2}\right)^1 = 0.0098$ ${}_{10}C_{10} \left(\frac{1}{2}\right)^{10} \left(\frac{1}{2}\right)^0 = 0.0010$ 0.0547 5.47%	10. Carol randomly chooses a day of the week six times. What is the probability of choosing Tuesday or Thursday at most three times? ${}_{6}C_3 \left(\frac{2}{7}\right)^3 \left(\frac{5}{7}\right)^3 = 0.1700$ ${}_{6}C_2 \left(\frac{2}{7}\right)^2 \left(\frac{5}{7}\right)^4 = 0.3187$ ${}_{6}C_1 \left(\frac{2}{7}\right)^1 \left(\frac{5}{7}\right)^5 = 0.3187$ ${}_{6}C_0 \left(\frac{2}{7}\right)^0 \left(\frac{5}{7}\right)^6 = 0.1328$.9402 94.02%
	11. The soda company prints prizes inside their bottle caps. The probability of winning a prize is 1 in 3. Out of eight bottle caps, what is the probability that at least five are winners? ${}_{8}C_5 \left(\frac{1}{3}\right)^5 \left(\frac{2}{3}\right)^3 = 0.0683$ ${}_{8}C_6 \left(\frac{1}{3}\right)^6 \left(\frac{2}{3}\right)^2 = 0.0171$ ${}_{8}C_7 \left(\frac{1}{3}\right)^7 \left(\frac{2}{3}\right)^1 = 0.0024$ ${}_{8}C_8 \left(\frac{1}{3}\right)^8 \left(\frac{2}{3}\right)^0 = 0.0002$ 0.088 8.8%	12. There are 7 boys and 5 girls on the debate team. Each week, one is chosen at random to lead the debate. For the next 10 weeks, what is the probability that no more than four girls are chosen to lead? ${}_{10}C_0 \left(\frac{5}{12}\right)^0 \left(\frac{7}{12}\right)^{10} = 0.0046$ ${}_{10}C_1 \left(\frac{5}{12}\right)^1 \left(\frac{7}{12}\right)^9 = 0.0326$ ${}_{10}C_2 \left(\frac{5}{12}\right)^2 \left(\frac{7}{12}\right)^8 = 0.1047$ ${}_{10}C_3 \left(\frac{5}{12}\right)^3 \left(\frac{7}{12}\right)^7 = 0.1995$ ${}_{10}C_4 \left(\frac{5}{12}\right)^4 \left(\frac{7}{12}\right)^6 = 0.2494$ 0.5908 59.08%

Name: _____

Unit 11: Probability & Statistics

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Homework 4: Binomial Theorem & Binomial Probability

**** This is a 2-page document! ******Directions:** Expand each binomial using the binomial theorem.

1. $(a+b)^9$

$${}^9C_0 a^9 b^0 + {}^9C_1 a^8 b^1 + {}^9C_2 a^7 b^2 + {}^9C_3 a^6 b^3 + {}^9C_4 a^5 b^4 + {}^9C_5 a^4 b^5 + {}^9C_6 a^3 b^6 + {}^9C_7 a^2 b^7 + {}^9C_8 a^1 b^8 + {}^9C_9 a^0 b^9$$

$$= a^9 + 9a^8b + 36a^7b^2 + 84a^6b^3 + 126a^5b^4 + 126a^4b^5 + 84a^3b^6 + 36a^2b^7 + 9ab^8 + b^9$$

2. $(m-2)^8$

$${}^8C_0 m^8 (-2)^0 + {}^8C_1 m^7 (-2)^1 + {}^8C_2 m^6 (-2)^2 + {}^8C_3 m^5 (-2)^3 + {}^8C_4 m^4 (-2)^4 + {}^8C_5 m^3 (-2)^5 + {}^8C_6 m^2 (-2)^6 + {}^8C_7 m^1 (-2)^7 + {}^8C_8 m^0 (-2)^8$$

$$= m^8 - 16m^7 + 112m^6 - 448m^5 + 1120m^4 - 1792m^3 + 1792m^2 - 1024m + 256$$

3. $(3k-1)^6$

$${}^6C_0 (3k)^6 (-1)^0 + {}^6C_1 (3k)^5 (-1)^1 + {}^6C_2 (3k)^4 (-1)^2 + {}^6C_3 (3k)^3 (-1)^3 + {}^6C_4 (3k)^2 (-1)^4 + {}^6C_5 (3k)^1 (-1)^5 + {}^6C_6 (3k)^0 (-1)^6$$

$$= 729k^6 - 1458k^5 + 1215k^4 - 540k^3 + 135k^2 - 18k + 1$$

4. $(2r+5s)^5$

$${}^5C_0 (2r)^5 (5s)^0 + {}^5C_1 (2r)^4 (5s)^1 + {}^5C_2 (2r)^3 (5s)^2 + {}^5C_3 (2r)^2 (5s)^3 + {}^5C_4 (2r)^1 (5s)^4 + {}^5C_5 (2r)^0 (5s)^5$$

$$= 32r^5 + 400r^4s + 2000r^3s^2 + 5000r^2s^3 + 6250rs^4 + 3125s^5$$

Directions: Find each binomial probability.

5. The probability that Ashley wins a race against Madelyn is 25%. What is the probability that Ashley wins **exactly** three of the next five races against Madelyn?

$$n=5, p=.25, q=.75, r=3$$

$${}^5C_3 (.25)^3 (.75)^2$$

$$= 0.0880$$

$$8.8\%$$

6. A coin is tossed 14 times. What is the probability of getting heads **exactly** 5 times?

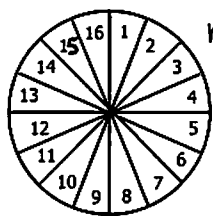
$$n=14, p=1/2, q=1/2, r=5$$

$${}^{14}C_5 (1/2)^5 (1/2)^9$$

$$= 0.122$$

$$12.22\%$$

7. The spinner below is divided into 16 equally spaced sections. If the spinner is spun 20 times, what is the probability of getting a perfect square **exactly** seven times?



$$n=20, p=\frac{4}{16}, q=\frac{12}{16}$$

$$r=7$$

$$20C_7 \left(\frac{1}{4}\right)^7 \left(\frac{3}{4}\right)^{13}$$

$$= 0.1124$$

$$11.24\%$$

8. The probability of hitting a target is $\frac{1}{8}$. What is the probability of hitting the target **exactly** twice in five tries?

$$n=5, p=\frac{1}{8}, q=\frac{7}{8}, r=2$$

$$5C_2 \left(\frac{1}{8}\right)^2 \left(\frac{7}{8}\right)^3$$

$$= 0.1047$$

$$10.47\%$$

9. A die is rolled eight times. What is the probability of getting **exactly** three 6's?

$$n=8, p=\frac{1}{6}, q=\frac{5}{6}, r=3$$

$$8C_3 \left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right)^5$$

$$= 0.1042$$

$$10.42\%$$

10. There are five blue crayons, seven yellow crayons, and eight red crayons in a box. If one is randomly drawn and replaced 15 times, find the probability of drawing **exactly** four blue crayons.

$$n=15, p=\frac{5}{20}, q=\frac{15}{20}, r=4$$

$$15C_4 \left(\frac{1}{4}\right)^4 \left(\frac{3}{4}\right)^{11}$$

$$= 0.2252$$

$$22.52\%$$

11. In hockey, Ed makes 7 goals for every 10 shots. If he takes 6 shots, what is the probability that he will make **at least** 5 goals?

$$n=6, p=\frac{7}{10}, q=\frac{3}{10}, r=5, 6$$

$$6C_5 \left(\frac{7}{10}\right)^5 \left(\frac{3}{10}\right)^1 = 0.3025$$

$$6C_6 \left(\frac{7}{10}\right)^6 \left(\frac{3}{10}\right)^0 = 0.1176$$

$$42.01\%$$

$$0.4201$$

12. The probability that it will snow on the last day of January is 85%. If the probability remains the same for the first eight days of February, what is the probability that it will snow **at least** five of those days in February?

$$n=8, p=0.85, q=0.15, r=5, 6, 7, 8$$

$$8C_5 (.85)^5 (.15)^3 = 0.0839$$

$$8C_6 (.85)^6 (.15)^2 = 0.2376$$

$$8C_7 (.85)^7 (.15)^1 = 0.3847$$

$$8C_8 (.85)^8 (.15)^0 = 0.2725$$

$$0.9787$$

$$97.87\%$$

13. Whenever Nicole rents a movie from iTunes, the probability that it will be a comedy is 52%. Of the next seven movies she rents, what is the probability that she rents **no more than** two comedies?

$$n=7, p=0.52, q=0.48, r=0, 1, 2$$

$$7C_0 (0.52)^0 (0.48)^7 = 0.0059$$

$$7C_1 (0.52)^1 (0.48)^6 = 0.0445$$

$$7C_2 (0.52)^2 (0.48)^5 = 0.1447$$

$$19.51\%$$

$$0.1951$$

14. The chance of winning a certain game at a carnival is 2 in 5. If Andy plays the game 12 times, what is the probability that he loses **at most** 3 times?

$$n=12, p=\frac{3}{5}, q=\frac{2}{5}, r=0, 1, 2, 3$$

$$12C_0 \left(\frac{3}{5}\right)^0 \left(\frac{2}{5}\right)^{12} = 0.00002$$

$$12C_1 \left(\frac{3}{5}\right)^1 \left(\frac{2}{5}\right)^{11} = 0.0003$$

$$12C_2 \left(\frac{3}{5}\right)^2 \left(\frac{2}{5}\right)^{10} = 0.0025$$

$$12C_3 \left(\frac{3}{5}\right)^3 \left(\frac{2}{5}\right)^9 = 0.0125$$

$$0.0153$$

$$1.53\%$$

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Unit 11: Probability & Statistics

Quiz 11-1: Probability**Part I – Counting Principle, Permutations, and Combinations**

1. A six-sided number cube is rolled and a card is drawn from a standard deck. Find the total number of outcomes.

$$6 \cdot 52$$

1. 312

2. A science quiz has eight multiple choice questions with five choices for each. Find the total number of ways to answer the questions.

$$5^8$$

2. 390,625

3. The golf club can send a team of four students to the next golf tournament. If there are 15 students in the club, how many different teams of four can be selected?

$$15C_4$$

3. C; 1365

4. There are 18 skaters competing in the competition. How many ways can they win the gold, silver, and bronze medal?

$$18P_3$$

4. P; 4896

5. What is the total number of different 10-letter arrangements that can be formed using the letters in the word **SUSPICIOUS**?

$$\frac{10!}{3!2!2!}$$

5. P; 151,200

Part II – Probability: Give each answer as fraction in simplest form.

6. What is the probability of drawing a jack or a red card from a standard deck of playing cards?

$$\frac{28}{52}$$

6. $\frac{7}{13}$

7. A bag contains 40 slips of paper numbered 1-40. If one is randomly chosen, what is the probability of not choosing a multiple of 3?

$$1 - \frac{13}{40}$$

7. $\frac{27}{40}$

8. A coin is tossed two times, then a month of the year is randomly select. What is the probability of getting tails each time, and a month that starts with the letter J?

$$\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{3}{12}$$

8. $\frac{1}{16}$

9. A bookshelf contains 9 mysteries, 7 biographies, and 4 science fiction books. A book is randomly selected, not replaced, then another is selected. Find the probability of selecting a biography then a mystery.

$$\frac{7}{20} \cdot \frac{6}{19}$$

9. $\frac{63}{380}$

10. A shape is randomly selected from the following quadrilaterals: parallelogram, rhombus, rectangle, square, and trapezoid. What is the probability that it has four right angles, given that it has four congruent sides?

10. $\frac{1}{2}$

11. Of the 100 students enrolled at the dance studio, 50 take ballet and 24 take tap dancing. Eight of the students who take ballet also take tap dancing. If a student at the studio is chosen at random, find the probability that they take ballet, if it is known that they do not take tap dancing.

$$11. \quad \frac{21}{38}$$

$$\frac{50-8}{76} = \frac{42}{76} = \frac{21}{38}$$

12. The table below represents the number of NFL teams that won or lost their first game of the season. Find the joint relative and marginal relative frequencies.

	Conference	
	AFC	NFC
Won	10	6
Lost	6	10

	Conference		
	AFC	NFC	Total
Won	0.3125	0.1875	0.5
Lost	0.1875	0.3125	0.5
Total	0.5	0.5	1

Use the tables above to find each probability.

13. $P(\text{AFC or NFC and won})$

$$13. \quad 50\%$$

$$14. \quad P(\text{NFC} \mid \text{won}) = \frac{0.1875}{0.5} = 37.5$$

$$14. \quad 37.5\%$$

$$15. \quad P(\text{lost} \mid \text{AFC}) = \frac{0.1875}{0.5} = 37.5$$

$$15. \quad 37.5\%$$

Part III – Binomial Probability: Give each answer as percent rounded to the nearest hundredth.

16. A coin is tossed 12 times. What is the probability of getting heads exactly 7 times?

$$16. \quad 19.34\%$$

$${}_{12}C_7 \left(\frac{1}{2}\right)^7 \left(\frac{1}{2}\right)^5 = 0.1934$$

17. Northern Airlines has a reputation for being on time 86% of the time. Find the probability that a Northern Airlines plane will be on time for at least four of its next five flights.

$$17. \quad 85.33\%$$

$$\left. \begin{aligned} {}_5C_4 (.86)^4 (.14)^1 &= .3829 \\ {}_5C_5 (.86)^5 (.14)^0 &= .4704 \end{aligned} \right\} 0.8533$$

18. Scott strikes out at bat 36% of the time. Find the probability that he strikes out no more than three times on his next eight times at bat.

$$18. \quad 68.47\%$$

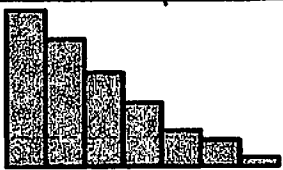
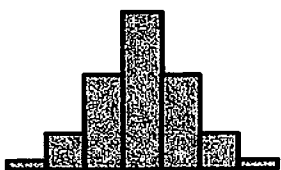
$$\left. \begin{aligned} {}_8C_0 (.36)^0 (.64)^8 &= .0281 \\ {}_8C_1 (.36)^1 (.64)^7 &= .1267 \\ {}_8C_2 (.36)^2 (.64)^6 &= .2494 \\ {}_8C_3 (.36)^3 (.64)^5 &= .2805 \end{aligned} \right\} 0.6847$$

Variance (σ^2)	The average squared distance of each value to the mean.	$\sigma^2 = \frac{\sum_{i=1}^n (x_i - \mu)^2}{n}$
Standard Deviation (σ)	The square root of the variance. * Most useful measure of dispersion.	$\sigma = \sqrt{\frac{\sum_{i=1}^n (x_i - \mu)^2}{n}}$
What do the numbers mean?	- A lower standard deviation indicates that <u>the values are closer together.</u> - A higher standard deviation indicates that <u>the values are more spread out.</u>	
Examples	Directions: Find the variance and standard deviation for each data set. 5. {85, 88, 91, 96, 100} mean = 92 $\sigma^2 = \frac{(85-92)^2 + (88-92)^2 + (91-92)^2 + (96-92)^2 + (100-92)^2}{5}$ $= \frac{(-7)^2 + (-4)^2 + (-1)^2 + (4)^2 + (8)^2}{5}$ $= \frac{49 + 16 + 1 + 16 + 64}{5}$ Variance (σ^2) = <u>29.2</u> Standard Deviation (σ) = <u>5.40</u>	
Calculator Steps: - Hit STAT, ENTER - Enter the data values in L1 - Hit STAT - Arrow over to CALC - Choose #1:1-Var Stats - Hit ENTER - Scroll down to find σ - Square to find σ^2 	6. {3, 4, 7, 8, 11, 12, 12, 15, 19} Variance (σ^2) = <u>23.62</u> Standard Deviation (σ) = <u>4.86</u>	
	7. {52, 47, 60, 55, 58, 53, 59, 63} Variance (σ^2) = <u>23.14</u> Standard Deviation (σ) = <u>4.81</u>	
	8. {127, 81, 135, 100, 97, 115, 72, 152, 63, 164} Variance (σ^2) = <u>1038.13</u> Standard Deviation (σ) = <u>32.22</u>	
	9. {25, 27, 21, 32, 35, 29, 23, 26, 24, 25, 28, 31} Variance (σ^2) = <u>14.98</u> Standard Deviation (σ) = <u>3.87</u>	

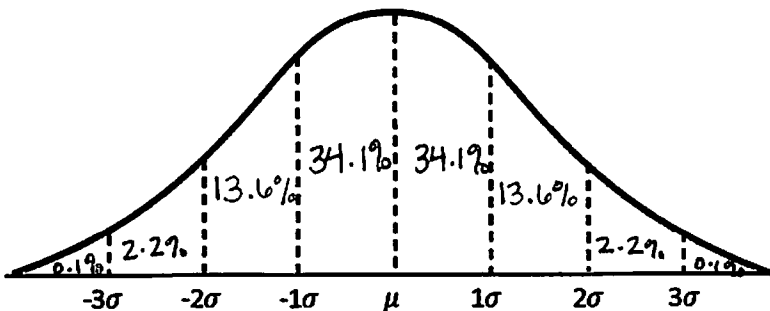
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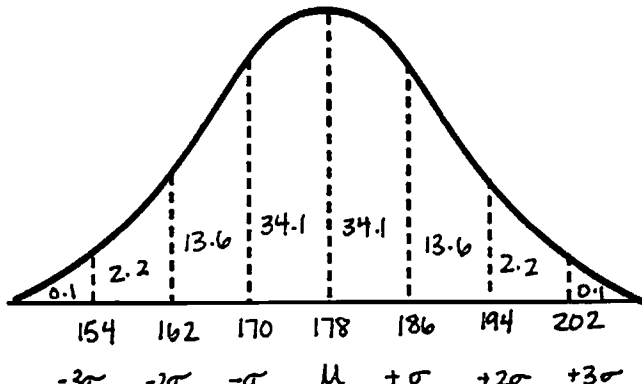
Main Ideas/Questions	Notes/Examples
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Data Distribution	The way data values are clustered.  Positively skewed  Negatively skewed  "Normal" Distribution
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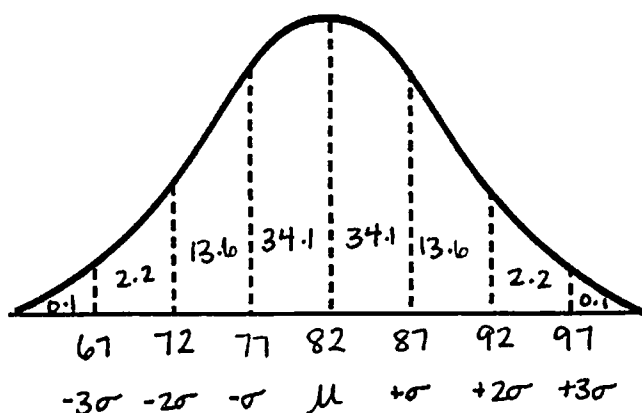
Normal Distribution	- A symmetrical <u>bell-shaped curve</u> defined by the mean (μ) and the standard deviation (σ) of the data set. - The area under the curve represents <u>probability</u>, with the total area equal to <u>1</u>.
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The Empirical Rule	In a normal distribution with mean μ and standard deviation σ: - Approximately <u>68.2%</u> of the data falls within 1σ of the mean. - Approximately <u>95.4%</u> of the data falls within 2σ of the mean. - Approximately <u>99.8%</u> of the data falls within 3σ of the mean. 
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Directions: Draw and label normal distribution curves, then answer the questions.

1. The weights of the 50 football players are normally distributed with a mean of 178 pounds and a standard deviation of 8 pounds. 	a) What percent of the players weigh between 178 lbs and 194 lbs? 47.7% b) What is the probability that a player weighs at most 170 lbs? 15.9% c) What is the probability that a player weighs less than 162 lbs or greater than 194 lbs? 4.6% d) How many players weigh between 170 lbs and 186 lbs? 68.2% of 50 = 34 players
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2. A set of 120 test scores are normally distributed with a mean of 82 and a standard deviation of 5.



- a) What percent of the scores are between 72 and 87?

81.8%

- b) What is the probability that a score is greater than 77?

84.3%

- c) What is the probability that a score is less than 82 or greater than 92?

$$< 82 = 50\%$$

$$> 92 = 2.3\%$$

52.3%

- d) About how many students scored outside two standard deviations of the mean?

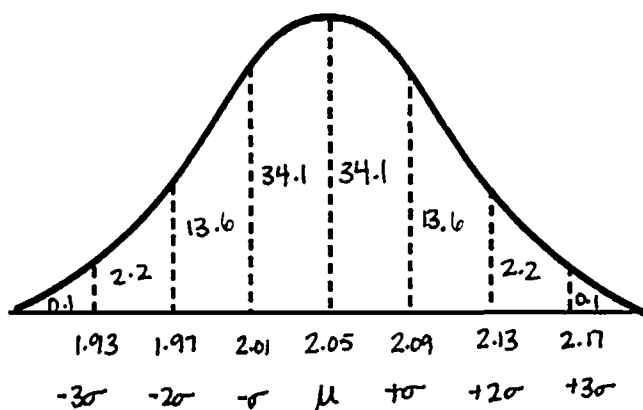
$$> 92 = 2.3\%$$

$$< 72 = 2.3\%$$

$$4.6\% \text{ of } 120 = 5.52$$

About 5 students

3. The price of a gallon of regular gasoline at 75 gas stations across the state is normally distributed with a mean of \$2.05 and a standard deviation of 4¢.



- a) What percent of gas stations sell a gallon of regular gas for less than \$1.97?

2.3%

- b) What percent of gas stations sell a gallon of regular gas for at least \$2.17?

0.1%

- c) What is the probability that a gas station sells a gallon of regular gas for less than \$1.97 or greater than \$2.05?

$$< 1.97 = 2.3\%$$

$$> 2.05 = 50\%$$

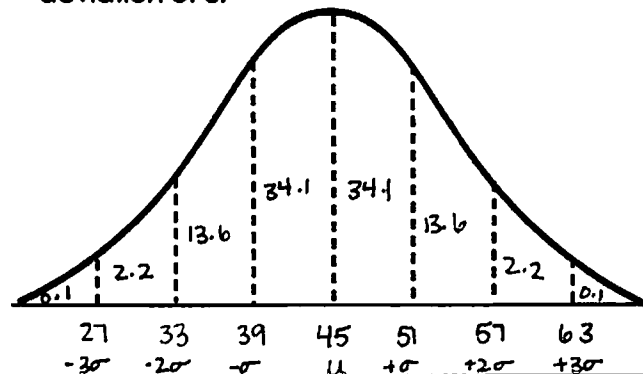
52.3%

- d) About how many stations sell a gallon of regular gas for no more than \$2.01?

$$15.9\% \text{ of } 75 = 11.93$$

About 11 gas stations

4. Mrs. Fuller recently tested her 120 keyboarding students to see how many words per minute they can type. The results were normally distributed with a mean of 45 and a standard deviation of 6.



- a) About how many students can type at least 39 words per minute?

$$> 39 = 84.1\%$$

$$84.1\% \text{ of } 120 = 100.92$$

About 100 students

- b) About how many students can type within one standard deviation of the mean?

$$68.2\% \text{ of } 120 = 81.84$$

About 81 students

- c) Students need to be in the top 2% in order to be eligible for the national typing competition. If Carla can type 56 wpm, is she eligible?

No, she is not eligible.

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Unit 11: Probability & Statistics

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Homework 5: Measures of Center, Variation, and Normal Distribution

**** This is a 2-page document! ******Directions:** Find the mean, median, and mode of each data set below.

1. {11, 14, 11, 5, 17, 28, 3}

3, 5, 11, 11, 14, 17, 28

Mean = $\frac{12.71}{1}$

Median = $\frac{11}{1}$

Mode(s) = $\frac{11}{1}$

2. {24, 29, 31, 16, 49, 52, 29, 35, 62, 29}

16, 24, 29, 29, 29, 31, 35, 49, 52, 62

Mean = $\frac{35.6}{1}$

Median = $\frac{30}{1}$

Mode(s) = $\frac{29}{1}$

Directions: Find the mean absolute deviation, variance, and standard deviation for each data set.

3. The following data shows the number of fish caught by a seven boy scouts on their camping trip:

{1, 2, 2, 4, 5, 6, 8} Mean = 4

$$\frac{3 + 2 + 2 + 0 + 1 + 2 + 4}{7}$$

MAD = $\frac{2}{1}$

Variance: $\sigma^2 = \frac{5.43}{1}$

Standard Deviation: $\sigma = \frac{2.33}{1}$

4. The following data shows the points scored by a football team during their first ten games of the season: {27, 32, 41, 9, 14, 20, 31, 33, 20, 13} Mean = 24

$$\frac{3 + 8 + 17 + 15 + 10 + 4 + 7 + 9 + 4 + 11}{10}$$

MAD = $\frac{8.8}{1}$

Variance: $\sigma^2 = \frac{97.02}{1}$

Standard Deviation: $\sigma = \frac{9.85}{1}$

5. The following data shows the high temperature for the past eight days:

{66, 46, 53, 50, 52, 47, 45, 49} mean = 51

$$\frac{15 + 5 + 2 + 1 + 1 + 4 + 6 + 2}{8}$$

MAD = $\frac{4.5}{1}$

Variance: $\sigma^2 = \frac{38.94}{1}$

Standard Deviation: $\sigma = \frac{6.24}{1}$

6. The following data shows the price of six different jars of pasta sauce at the grocery store:

{\$2.79, \$1.99, \$4.29, \$2.49, \$2.29, \$3.49} mean = 2.89

$$\frac{.10 + .90 + 1.40 + .40 + .60 + .60}{6}$$

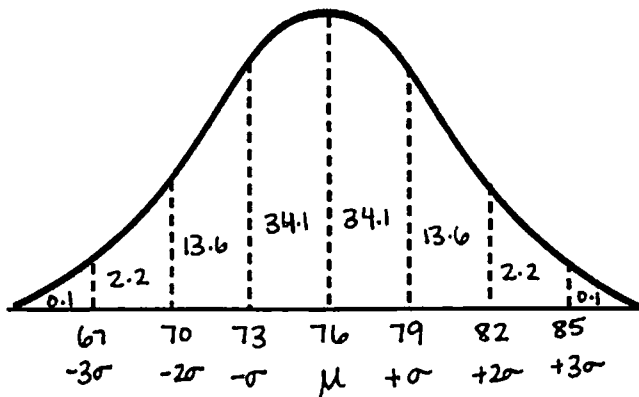
MAD = $\frac{0.67}{1}$

Variance: $\sigma^2 = \frac{0.61}{1}$

Standard Deviation: $\sigma = \frac{0.78}{1}$

For questions 7 and 8, draw the normal distribution curve, then answer the questions.

7. A set of 125 golf scores are normally distributed with a mean of 76 and a standard deviation of 3.



- a) What percent of the scores are between 67 and 85?

99.8%

- b) What is the probability that a score is no more than 79?

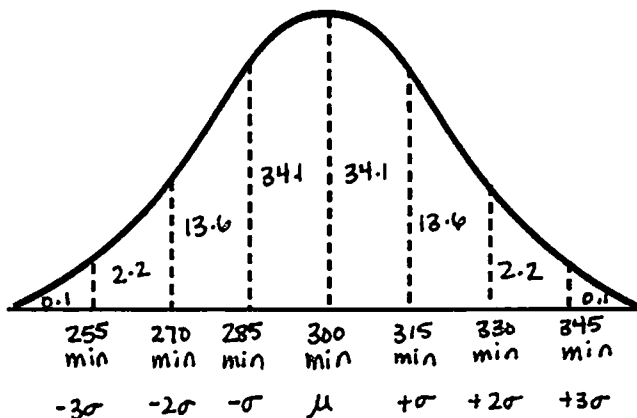
84.1%

- c) About how many scores fell between one standard deviation of the mean?

68.2% of 125 = 85.25

About 85 scores

8. The talk-time battery life of a group of cell phones is normally distributed with a mean of 5 hours and a standard deviation of 15 minutes.



- a) What percent of the phones have a battery life of at least 4 hours and 45 minutes? (285 min)

84.1%

- b) What percent of the phones have a battery life between 4.5 hours and 5.25 hours? (270 min - 315 min)

81.8%

- c) What percent of the phones have a battery life less than 5 hours or greater than 5.5 hours?

< 300 min = 50

> 330 min = 2.3

52.3%

9. The number of hours that the employees at the grocery store worked last week is normally distributed with a mean of 24 and a standard deviation of 6. If there are 60 total employees, approximately how many worked at least 30 hours last week?

$\mu = 24$

15.9% of 60 = 9.54

$+1\sigma = 30$

About 9 employees

10. The grade point average (GPA) of the students at Lakeview High School is normally distributed with a mean of 3.1 and a standard deviation of 0.3. If there are 1800 students enrolled at the school, approximately how many have a GPA between 2.5 and 3.7?

2.5 ← 2.8 ← 3.1 → 3.4 → 3.7
-2σ -σ μ +σ +2σ

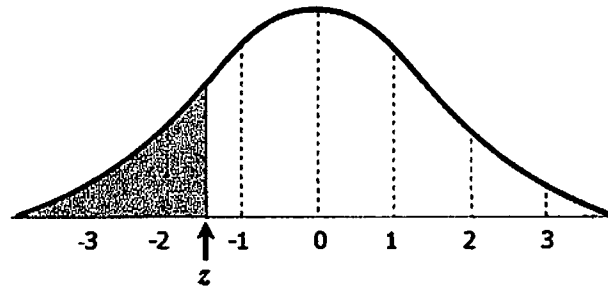
95.4%

Name:	Date:
Topic:	Class:

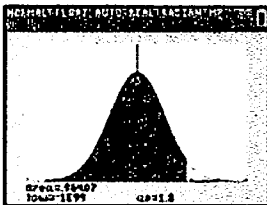
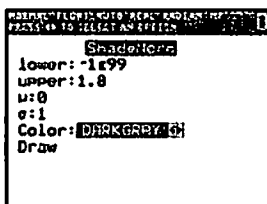
Main Ideas/Questions	Notes/Examples
Z-SCORES	- A value that shows how many <u>standard deviations</u> a data value is from the <u>mean</u>. - When the z-score is positive, the data value is <u>above</u> the mean. - When the z-score is negative, the data value is <u>below</u> the mean. - When $z = 0$, the data value <u>equal to</u> the mean
Z-Score Formula	To find the z-score for a data value in a set that is normally distributed, use: $z = \frac{X - \mu}{\sigma}$ X = the data value μ = the mean σ = the standard deviation
EXAMPLES	1. A radar detector records the speeds of a group of cars that pass by. If the mean is 46 mph and the standard deviation is 2.8 mph, find the z-scores for each data value. a) 52 $z = \frac{52 - 46}{2.8} = \boxed{2.14}$ b) 42 $z = \frac{42 - 46}{2.8} = \boxed{-1.43}$ c) 47 $z = \frac{47 - 46}{2.8} = \boxed{0.36}$ 2. The mean number of total miles ran last week by each member on the track team was 4 with a standard deviation of 1.2. If Clay's z-score was 2.5, how many miles did he run? $2.5 = \frac{x - 4}{1.2}$ $3 = x - 4$ $7 = x$ 7 miles 3. A set test papers is normally distributed with a standard deviation of 5. If Riley scored an 83 with a z-score of -0.4, what was the mean? $-0.4 = \frac{83 - \mu}{5}$ $-2 = 83 - \mu$ $85 = \mu$ Mean was 85 4. The heights of a group of trees is normally distributed with a mean of 14.3 feet. If the z-score for a 20-foot tall tree is 1.9, what is the standard deviation? $1.9 = \frac{20 - 14.3}{\sigma}$ $\sigma = \frac{20 - 14.3}{1.9}$ $\sigma = 3$

Standard Normal Distribution

- A normal distribution in which $\mu = 0$ and $\sigma = 1$.
- The distribution is "standardized" by the z-score.



Calculator Directions



*

$$-\infty = -1E99$$

$$\infty = 1E99$$

First, change your window dimensions:

Xmin: -4, Xmax: 4, Ymin: 0, Ymax: 0.5

To graph the curve and find the probability:

- Step 1: Find the z-score for each data value.
- Step 2: Hit 2nd -> VARS
- Step 3: Arrow over to DRAW
- Step 4: Select 1:ShadeNorm(
- Step 5: Enter the lower and upper bounds for the z-scores
- Step 6: Scroll down and hit DRAW

Example: The golf scores in a tournament is normally distributed with a mean of 76 and a standard deviation of 2.6. Find each probability.

5. $P(z < 2.1)$

$$.982136$$

$$\boxed{98.21\%}$$

6. $P(X < 74)$

$$z = \frac{74 - 76}{2.6} = -0.77$$

$$.22065$$

$$\boxed{22.07\%}$$

7. $P(z > -1.5)$

$$.933193$$

$$\boxed{93.32\%}$$

8. $P(X > 81)$

$$z = \frac{81 - 76}{2.6} = 1.92$$

$$.027429$$

$$\boxed{2.74\%}$$

9. $P(-0.2 < z < 1.8)$

$$.543329$$

$$\boxed{54.33\%}$$

10. $P(77 < X < 83)$

$$z = \frac{77 - 76}{2.6} = 0.38$$

$$z = \frac{83 - 76}{2.6} = 2.69$$

$$.3484$$

$$\boxed{34.84\%}$$

Name: _____

Unit 11: Probability & Statistics

Date: _____ Bell: _____

Homework 6: Z-Scores & Standard Normal Distribution

**** This is a 2-page document! ****

1. The number of calories burned at the gym is normally distributed with a mean of 425 and a standard deviation of 51. Find the z-scores for each data value.

a) 268

$$\frac{268 - 425}{51} = -3.08$$

b) 512

$$\frac{512 - 425}{51} = 1.71$$

c) 450

$$\frac{450 - 425}{51} = 0.49$$

2. The air pressure of each tire in a group is normally distributed with a mean of 39.2 pounds per square inch (psi) and a standard deviation of 2.5 psi. If a tire in this group has a z-score of -1.8, find its air pressure.

$$-1.8 = \frac{X - 39.2}{2.5}$$

$$-4.5 = X - 39.2$$

$$\boxed{34.7 = X}$$

3. A set of math tests is normally distributed with a mean of 81 and a standard deviation of 5. If Adam's z-score was -0.6 and Leah's z-score was 2.2, how many points higher did Leah score than Adam?

$$A: -0.6 = \frac{X - 81}{5}$$

$$-3 = X - 81$$

$$78 = X$$

$$L: 2.2 = \frac{X - 81}{5}$$

$$11 = X - 81$$

$$92 = X$$

$$\boxed{14 \text{ points}}$$

4. The prices of various cereal boxes is normally distributed with a standard deviation of 0.5. If a box of Lucky Charms sells for \$4.29 and has a z-score of 1.6, what is the mean price of a box of cereal?

$$1.6 = \frac{4.29 - \mu}{0.5}$$

$$0.8 = 4.29 - \mu$$

$$\boxed{3.49 = \mu}$$

5. The number of homes sold each year by a realtor is normally distributed with a mean of 54. If the realtor sold 36 homes last year with a z-score of -2.4, what is the standard deviation?

$$-2.4 = \frac{36 - 54}{\sigma}$$

$$\sigma = \frac{36 - 54}{-2.4}$$

$$\boxed{\sigma = 7.5}$$

6. The last math exam of the year was normally distributed with a mean of 85 and a standard deviation of 5. Xavier needed at least a 95 on this test to get an A for the year. His teacher gave him his z-score: 1.8. Did Xavier get an A for the year?

$$1.8 = \frac{X - 85}{5}$$

$$9 = X - 85$$

$$94 = X$$

No he did not.

7. The 400-meter race times recorded in the boys track meet was normally distributed with a standard deviation of 3 seconds. If David finished the race in 51.2 seconds with a z-score of -2.6, what was the mean time?

$$-2.6 = \frac{51.2 - \mu}{3}$$

$$-7.8 = 51.2 - \mu$$

$$\boxed{59 = \mu}$$

Use for questions 8-14: The weights of dogs in a dog show is normally distributed with a mean of 58 pounds and a standard deviation of 17.2 pounds. Use a standard normal distribution curve to find each probability.

8. $P(z < -0.8)$

0.211855

21.19%

9. $P(z > 1.32)$

0.093418

9.34%

10. $P(-2.8 < z < -1.17)$

.118445

11.84%

11. $P(X > 51)$

$\frac{51-58}{17.2} = -0.41$

65.91%

12. $P(35 < X < 62)$

$\frac{35-58}{17.2} = -1.34$ $\frac{62-58}{17.2} = 0.23$

50.08%

13. $P(X < 102)$

$\frac{102-58}{17.2} = 2.56$

99.48%

14. $P(X > 85)$

$\frac{85-58}{17.2} = 1.57$

5.82%

15. The high temperatures in Virginia Beach during the month of July are normally distributed with a mean of 83° and a standard deviation of 6°. Find the probability that on a given day during the month, the temperature is greater than 92°.

$\frac{92-83}{6} = 1.5$

$z > 1.5$

6.68%

16. The mean number of hours that the average person watches television each day is 4.18 hours with a standard deviation of 1.19 hours. Find the probability that someone watches between 3 and 5 hours per day.

$\frac{3-4.18}{1.19} = -0.99$

$\frac{5-4.18}{1.19} = 0.69$

$-0.99 < z < 0.69$

59.38%

17. The mean song length on Jack's iPad is 3 minutes and 3 seconds, with a standard deviation of 24 seconds. Find the probability that the next song to play is no more than 3 minutes and 45 seconds.

mean = 183 sec

$\frac{225-183}{24} = 1.75$

$z < 1.75$

95.99%

Name: _____

Algebra 2

Date: _____ Per: _____

Unit 11: Probability & Statistics

Quiz 11-2: Statistics

1. The following data represents the number of calories in seven sandwiches at the restaurant: {740, 610, 490, 370, 500, 540, 460}. Find the mean absolute deviation. Round your answer to the nearest hundredth.

1. 85.71 cal.

$$\mu = 530$$

$$\frac{210 + 80 + 40 + 160 + 30 + 10 + 70}{7} = \frac{600}{7}$$

2. The speed in miles per hour of each of Sam's last ten pitches were recorded as follows: {82, 78, 83, 80, 75, 82, 79, 80, 83, 81}. Find the variance. Round your answer to the nearest hundredth.

2. 5.62 mph

$$\sigma = 2.37$$

3. The following data below represents the running times of six movies. Find the standard deviation of the running times. Give your answer to the nearest hundredth of a minute.

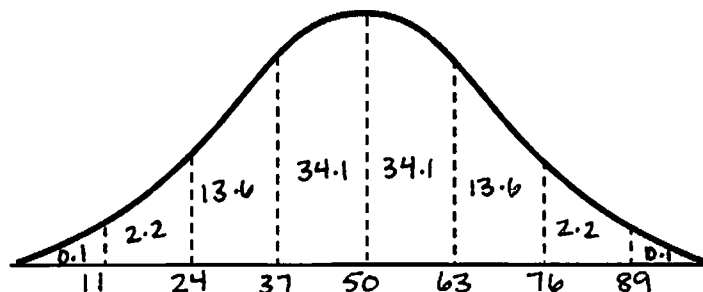
3. 29.92 min

Movie	A	B	C	D	E	F
Running Time	1 hr 47 min	1 hr 32 min	2 hr 14 min	1 hr 56 min	2 hr 32 min	3 hr 2 min
	107	92	134	116	152	182

4. The gas mileage for a group of cars is normally distributed with a mean of 24 miles per gallon and a variance of 34.81. What is the standard deviation?

4. 5.9 mpg

Use for questions 5-8: The number of days a group of 250 homes is on the market is normally distributed with a mean 50 and a standard deviation of 13. Label the normal distribution curve, then answer the questions.



5. What percent of the homes are on the market between 11 and 89 days?

5. 99.8%

6. What is the probability that a home is on the market for at least 37 days?

6. 84.3%

7. Approximately how many homes were on the market between 37 and 63 days?

$$68.2\%$$

7. 170 homes

8. Approximately how many homes were on the market for no more than 24 days?

$$2.3\%$$

8. 5 homes

9. The number of text messages that teenagers send per month is normally distributed with a mean of 3,400 and a standard deviation of 450. If Kendra sent 4,415 text messages last month, find her z-score to the nearest hundredth.

$$\frac{4415 - 3400}{450}$$

9. 2.26

10. The daily high temperatures in Philadelphia for the first 15 days in February is normally distributed with a standard deviation of 2.5°. If it was 37° on Valentine's Day with a z-score of 2.4, find the mean temperature.

$$2.4 = \frac{37 - \mu}{2.5}$$

$$31 = \mu$$

10. 31°

11. The price of a gallon of milk at various stores is normally distributed with a mean of \$2.89 and a standard deviation of \$0.2. The price of a gallon of milk at SmartShop has a z-score of -1.3 while the price of a gallon of milk at Wegmans has a z-score of 0.6. How much more is a gallon of milk at Wegmans compared to SmartShop?

$$SS: -1.3 = \frac{x - 2.89}{0.2}$$

$$W: 0.6 = \frac{x - 2.89}{0.2}$$

$$2.63 = x$$

$$3.01 = x$$

11. \$0.38

12. Mark recently took a road trip across the country. The number of miles he drove each day was normally distributed with a mean of 450. If he drove 431.8 miles on the last day with a z-score of -0.7, what is the standard deviation?

$$-0.7 = \frac{431.8 - 450}{\sigma}$$

$$\sigma = \frac{431.8 - 450}{-0.7}$$

12. 26

Use for questions 13-16: The number of points scored by each team in the NHL at the end of the season is normally distributed with a mean of 89 and a standard deviation of 11. Find each probability. (X represents a data value and z represents a z-score)

13. $P(z > 1.3)$

13. 9.68%

14. $P(X < 85) \quad \frac{85 - 89}{11} = -0.36$

$$z < -0.36$$

14. 35.94%

15. $P(71 < X < 96)$

$$\frac{71 - 89}{11} = -1.64$$

$$\frac{96 - 89}{11} = 0.64$$

$$-1.64 < z < 0.64$$

15. 68.84%

16. If there are 30 teams, approximately how many scored at least 76 points?

$$\frac{76 - 89}{11} = -1.18$$

$$z > -1.18 \rightarrow 88.1\%$$

16. 26 teams

PROBABILITY & STATISTICS *Formulas Reference*

Term	Definition	Formula
Fundamental Counting Principle	If decision M can be made "m" ways and decision N can be made "n" ways, then the two can be made "m · n" ways.	$m \cdot n$
Factorial	The product of natural numbers from "n" down to 1.	$n! = n \cdot (n-1) \cdots 3 \cdot 2 \cdot 1$
Permutation	An arrangement or line up where order matters.	$nPr = \frac{n!}{(n-r)!}$
Combination	An arrangement or group where order doesn't matter.	$nCr = \frac{n!}{r!(n-r)!}$
Independent Events	When the outcome of an event does not affect the outcome of another.	$P(A \text{ and } B) = P(A) \cdot P(B)$
Dependent Events	When the outcome of an event does affect the outcome of another.	$P(A \text{ and } B) = P(A) \cdot P(B, \text{ after } A \text{ occurred})$
Conditional Probability	Given events A and B, the probability of event B occurring knowing that A already occurred.	$P(B A) = \frac{P(A \text{ and } B)}{P(A)}$
Binominal Probability	The probability of two mutually exclusive events, each with a success and a failure.	$nCr \cdot p^r \cdot q^{n-r}$

Binomial
Theorem

The expansion of a binomial
without multiplying at length.

$$\sum_{k=0}^n {}^nC_k \cdot a^{n-k} \cdot b^k$$

Mean
Absolute
Deviation

The average distance of each
value to the mean.

$$\frac{\sum_{i=1}^n |x_i - \mu|}{n}$$

Variance

The average squared distance
of each value to the mean.

$$\sigma^2 = \frac{\sum_{i=1}^n (x_i - \mu)^2}{n}$$

Standard
Deviation

The square root of the
variance.

$$\sigma = \sqrt{\frac{\sum_{i=1}^n (x_i - \mu)^2}{n}}$$

z-score

A value that shows how
many standard deviations a
value is to the mean.

$$z = \frac{x - \mu}{\sigma}$$

Extra Notes:

Unit 11 Test Study Guide (Probability & Statistics)

Name: _____

Date: _____ Block: _____

Topic 1: Fundamental Counting Principle, Permutations, & Combinations

1. There are ten questions on Jack's science quiz: six multiple choice questions with four choices each and four true/false questions. How many ways can he answer the questions? $4^6 \cdot 2^4 = \boxed{65,536}$	2. The Sandwich Shop offers combo meals with a choice of one sandwich, one side, and one drink. In total, there are 560 combos available. If there are 16 sandwiches and 5 drinks available, how many sides are there? $16 \cdot 5 \cdot X = 560$ $80X = 560$ $X = \boxed{7}$
Determine whether the situation represents a permutation or combination, then solve.	
3. Marlena must answer three out of eight essay questions on her writing test. How many ways can she choose 3? $8C_3 = \boxed{56}$	4. The Grandview High School Band plans to play six pieces of music in their upcoming concert. How many ways can they arrange the order of their performance? $6! = \boxed{720}$
5. How many ways can a president, vice president, and treasurer be elected from a group of 40 students? $40P_3 = \boxed{59,280}$	6. Josh must submit 4 paintings as part of his application to art school. If he has 25 to choose from, how many ways can he pick 4? $25C_4 = \boxed{12,650}$
7. How many different 12-letter arrangements can be made using the letters in the word CARELESSNESS? $\frac{12!}{3! \cdot 4!} = \boxed{3,326,400}$	8. The field hockey coach chooses players to put away equipment after each practice. If there are 16 players on the team, how many ways can she choose five? $16C_5 = \boxed{4368}$

Topic 2: Theoretical Probability/Independent vs. Dependent Events

9. Kate is randomly choosing a date in the month of August. What is the probability that she chooses a two-digit number or a prime number? $\frac{26}{31}$	10. Erik is randomly choosing a card from a standard deck. What is the probability that he chooses a card that is red and a multiple of three? $\frac{6}{52} = \frac{3}{26}$
11. One of the 50 states is chosen at random. What is the probability of not choosing a state that begins with the letter N? $1 - \frac{8}{50} = \frac{21}{25}$	12. Tony's wallet contains six \$1 bills, four \$5 bills, two \$10 bills, and three \$20 bills. If Tony randomly removes a bill, what is the probability that it is at least \$5? $\frac{9}{15} = \frac{3}{5}$

13. A number 1-30 is randomly selected, followed by a letter from the word TARGET. What is the probability of choosing a perfect square then the letter T? $\frac{5}{30} \cdot \frac{2}{6} = \boxed{\frac{1}{18}}$	14. Carter is handing out Valentine's Day cards to his friends. He has 8 Batman cards, 12 Superman cards, and 5 Captain American cards in a bag. What is the probability that the first two he chooses are Batman? $\frac{8}{25} \cdot \frac{7}{24} = \boxed{\frac{7}{75}}$
Use for questions 15-16: A bag contains 25 red chips, 18 blue chips, and 29 yellow chips. A chip is selected, replaced, then another chip is chosen. Find each probability.	
15. $P(\text{blue then red})$ $\frac{18}{72} \cdot \frac{25}{72} = \boxed{\frac{25}{288}}$	16. $P(\text{yellow then blue})$ $\frac{29}{72} \cdot \frac{18}{72} = \boxed{\frac{29}{288}}$
Use for questions 17-18: A box of pizza contains three cheese slices, seven pepperoni slices, and five supreme slices. Mark randomly chooses a slice, eats it, then chooses another. Find each probability.	
17. $P(\text{cheese then supreme})$ $\frac{3}{15} \cdot \frac{5}{14} = \boxed{\frac{1}{14}}$	18. $P(\text{two pepperonis})$ $\frac{7}{15} \cdot \frac{6}{14} = \boxed{\frac{1}{5}}$

Topic 3: Conditional Probability

19. Mason randomly chooses a card from a standard deck of playing cards. What is the probability that it is not an ace, given it is a red card?

$$\frac{24}{26} = \boxed{\frac{12}{13}}$$

20. One of the 50 United States is randomly selected. What is the probability that it has two syllables, given it starts with the letter M?

$$\frac{0}{8} = \boxed{0}$$

21. Of the 318 sophomores, 140 are taking Algebra 2 and 102 are taking Chemistry. Twenty-six of those taking Algebra 2 are also taking chemistry. If a sophomore is chosen at random, find the probability that they are taking Algebra 2, if it is known that they do not take Chemistry.

$$\frac{114}{216} = \boxed{\frac{19}{36}}$$

22. A group of cell phone owners were asked whether they had an Android or apple phone. Use the results in the table to find each probability.

	Male	Female
Android	34	19
Apple	30	42

(125)

a) $P(\text{a male that owns an android})$

$$\boxed{\frac{34}{125}}$$

b) $P(\text{apple} \mid \text{male})$

$$\frac{30}{64} = \boxed{\frac{15}{32}}$$

c) $P(\text{female} \mid \text{android})$

$$\boxed{\frac{19}{53}}$$

23. The table below shows the number of gold, silver, and bronze medals won by the United States and China during the 2012 summer Olympics. Find the joint relative and marginal relative frequencies

	U.S.	China
Gold	46	38
Silver	28	28
Bronze	29	22

	U.S.	China	Total
Gold	.2408	.1990	.4398
Silver	.1466	.1466	.2932
Bronze	.1518	.1152	.2670
Total	.5392	.4608	1

(191)

If a medal from those above is randomly selected, find each probability.

a) $P(\text{a U.S. bronze medal})$

$$15.18\%$$

b) $P(\text{a gold or silver medal})$

$$73.3\%$$

c) $P(\text{silver} \mid \text{China})$

$$\frac{.1466}{.4608}$$

$$31.81\%$$

d) $P(\text{U.S.} \mid \text{gold})$

$$\frac{.2408}{.4398}$$

$$54.75\%$$

e) $P(\text{China} \mid \text{not silver})$

$$\frac{.342}{.7068}$$

$$44.45\%$$

f) $P(\text{bronze} \mid \text{not China})$

$$\frac{.1518}{.5392}$$

$$28.15\%$$

Topic 4: Binomial Probability & Binomial Theorem

Use for questions 24-27: The soda company is printing prizes on the inside of their bottle caps, with a 3 in 5 chance of winning. If Tom purchases a 12-pack of soda, find each probability.

24. $P(\text{exactly two prizes})$

$${}_{12}C_2 \left(\frac{3}{5}\right)^2 \left(\frac{2}{5}\right)^{10} = .0025$$

$$.25\%$$

25. $P(\text{exactly five prizes})$

$${}_{12}C_5 \left(\frac{3}{5}\right)^5 \left(\frac{2}{5}\right)^7 = .1009$$

$$10.09\%$$

26. $P(\text{at least ten prizes})$

$$\begin{aligned} & {}_{12}C_{10} \left(\frac{3}{5}\right)^{10} \left(\frac{2}{5}\right)^2 \\ & {}_{12}C_{11} \left(\frac{3}{5}\right)^{11} \left(\frac{2}{5}\right)^1 \\ & {}_{12}C_{12} \left(\frac{3}{5}\right)^{12} \left(\frac{2}{5}\right)^0 \\ & = .0835 \end{aligned}$$

$$8.35\%$$

27. $P(\text{no more than three prizes})$

$$\begin{aligned} & {}_{12}C_3 \left(\frac{3}{5}\right)^3 \left(\frac{2}{5}\right)^9 \\ & {}_{12}C_2 \left(\frac{3}{5}\right)^2 \left(\frac{2}{5}\right)^{10} \\ & {}_{12}C_1 \left(\frac{3}{5}\right)^1 \left(\frac{2}{5}\right)^{11} \\ & {}_{12}C_0 \left(\frac{3}{5}\right)^0 \left(\frac{2}{5}\right)^{12} \\ & = .0153 \end{aligned}$$

$$1.53\%$$

Expand each binomial using the binomial theorem.

28. $(a-2)^9$

$$\begin{aligned} & {}^9C_0(a)^9(-2)^0 + {}^9C_1(a)^8(-2)^1 + {}^9C_2(a)^7(-2)^2 + {}^9C_3(a)^6(-2)^3 + \\ & {}^9C_4(a)^5(-2)^4 + {}^9C_5(a)^4(-2)^5 + {}^9C_6(a)^3(-2)^6 + {}^9C_7(a)^2(-2)^7 \\ & + {}^9C_8(a)^1(-2)^8 + {}^9C_9(a)^0(-2)^9 \\ & = a^9 - 18a^8 + 144a^7 - 672a^6 + 2016a^5 - 4032a^4 + 5376a^3 - 4608a^2 + 2304a - 512 \end{aligned}$$

29. $(4k+1)^5$

$$\begin{aligned} & {}^5C_0(4k)^5(1)^0 + {}^5C_1(4k)^4(1)^1 + {}^5C_2(4k)^3(1)^2 + {}^5C_3(4k)^2(1)^3 \\ & + {}^5C_4(4k)^1(1)^4 + {}^5C_5(4k)^0(1)^5 \end{aligned}$$

$$1024k^5 + 1280k^4 + 640k^3 + 160k^2 + 20k + 1$$

Topic 5: Statistics: Measures of Variation, Normal Distribution, and z-Scores

Find the mean absolute deviation, variance, and standard deviation for each data set.

30. The following data represents the last ten field goal attempts, in yards, made by the kicker.

{54, 29, 33, 62, 45, 39, 41, 52, 59, 36} $\mu = 45$

$$\frac{9 + 16 + 12 + 17 + 0 + 6 + 4 + 7 + 14 + 9}{10}$$

MAD = 9.4

Variance: $\sigma^2 = \underline{114.70}$

Standard Deviation: $\sigma = \underline{10.71}$

31. The following data shows the grades of the Carl's last 8 quiz grades in Algebra 2.

{72, 93, 52, 86, 100, 81, 79, 61} $\mu = 78$

$$\frac{6 + 15 + 26 + 8 + 22 + 3 + 1 + 17}{8}$$

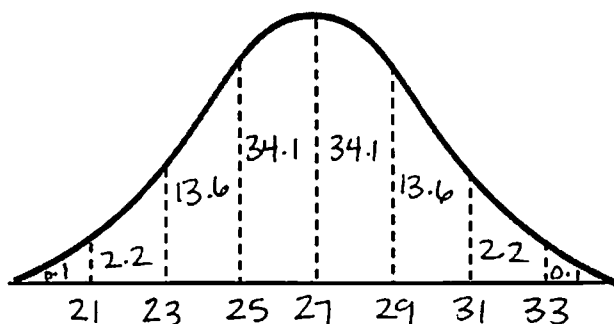
MAD = 12.25

Variance: $\sigma^2 = \underline{222.90}$

Standard Deviation: $\sigma = \underline{14.93}$

Label the normal distribution curve, then answer the questions that follow.

The ages of the 32 recruits in police academy are normally distributed with a mean of 27 with a standard deviation of 2.



32. What percent of the recruits are between ages 23 and 27?

47.7%

33. What is the probability that a recruit is at least 31 years old?

2.3%

34. Approximately how many recruits are at most 29 years old?

26 recruits

35. The blood pressure of a group of adults is normally distributed with a mean of 122 and a standard deviation of 13. If Mina's blood pressure is 108, find her z-score.

$$z = \frac{108 - 122}{13} = \boxed{-1.08}$$

36. The daily balance of a checking account over the course of a month is normally distributed with a standard deviation of \$84.10. If the balance in the account on a particular day was \$317.34 with a z-score of -2.4, find the mean balance.

$$-2.4 = \frac{317.34 - \mu}{84.10}$$

$$\boxed{519.18 = \mu}$$

37. The length of a certain fish species is normally distributed with a mean of 15 cm. If a fish in this species is 18.8 cm with a z-score of 1.9, what is the standard deviation?

$$1.9 = \frac{18.8 - 15}{\sigma}$$

$$\boxed{\sigma = 2}$$

38. The number of visits to the gym each year by the members at InFit is normally distributed with a mean of 97 and a standard deviation of 25. If Kurt's z-score is -1.8 and Mindy's z-score is 2.4, how many more times did Mindy visit the gym than Kurt?

$$-1.8 = \frac{x - 97}{25} \quad x = 52$$

$\boxed{105}$

$$2.4 = \frac{x - 97}{25} \quad x = 157$$

Use for questions 39-42: The shoe sizes of the 36 students in Samantha's PE class are normally distributed with a mean of 8.5 and a standard deviation of 1.5.

39. What percent of the students have a shoe size between 7 and 11?

$$\frac{7 - 8.5}{1.5} = -1$$

$\boxed{79.39\%}$

$$\frac{11 - 8.5}{1.5} = 1.67$$

40. What is the probability that a student will have a maximum shoe size of 9.5?

$$\frac{9.5 - 8.5}{1.5} = 0.67$$

$\boxed{74.86\%}$

41. Approximately how many students wear at least a size 6?

$$\frac{6 - 8.5}{1.5} = -1.67$$

$\boxed{95.25\%}$

42. Approximately how many students wear a shoe size between 8 and 10?

$$\frac{8 - 8.5}{1.5} = -0.33$$

47.06%

$$\frac{10 - 8.5}{1.5} = 1$$

$\boxed{17 \text{ students}}$

Name: _____

Date: _____ Bell: _____

Unit 11 Test

Probability & Statistics

1. The library labels each book with a five-digit code to identify the book in their computer system. Each code is 3 digits (0-9) followed by two letters. How many codes are possible? $10 \cdot 10 \cdot 10 \cdot 26 \cdot 26$ A. 468 B. 492,804 C. 527,000 D. 676,000 D	2. A test has 30 multiple choice questions with 4 choices each. Which expressions represents the total number of ways to answer the questions? A. 4^{30} B. 30^4 C. ${}_{30}P_4$ D. $4 \cdot 30$ A
3. A coach needs to choose two players from the hockey team to be captain and assistant captain. If there are 20 players on the team, how many ways he can assign the two roles? ${}_{20}P_2$ A. 100 B. 190 C. 380 D. 450 C	4. In a card game, each player is dealt a hand of 7 cards from a deck of 52 playing cards. Which expression represents the number of hands possible? A. $7! \cdot 52!$ B. $\frac{52!}{7!}$ C. ${}_{52}P_7$ D. ${}_{52}C_7$ D
5. SweetTooth Frozen Yogurt has 12 different toppings to choose from. If they add two new toppings, how many ways more ways can Ian choose three? ${}_{12}C_3 = 220$ ${}_{14}C_3 = 364$ A. 220 B. 144 C. 364 D. 864 B	6. How many different ways can the 11 letters in the word PROBABILITY be arranged? $\frac{11!}{2! \cdot 2!}$ A. 1,663,200 B. 7,241,600 C. 9,979,200 D. 39,916,800 C
7. A month of the year is selected at random. What is the probability that it has 31 days or starts with the letter J? $\frac{8}{12}$ A. $\frac{1}{6}$ B. $\frac{3}{4}$ C. $\frac{7}{12}$ D. $\frac{2}{3}$ D	8. A card is selected at random from a standard deck of 52 playing cards. What is the probability that it is not a face card? $\frac{40}{52}$ A. $\frac{3}{13}$ B. $\frac{10}{13}$ C. $\frac{4}{13}$ D. $\frac{9}{13}$ B

9. A die is rolled then a letter in the word STATISTICS is randomly selected. What is the probability of at least a three, then the letter S?

$$\frac{4}{6} \cdot \frac{3}{10}$$

A. $\frac{1}{5}$

C. $\frac{3}{20}$

B. $\frac{1}{4}$

D. $\frac{29}{30}$

A

10. Gavin and Jack are practicing shots against their goalie. On their last 15 attempts, Gavin made 6 and Jack made 7. Based on this performance, what is the probability that they both make their next shot?

A. $\frac{1}{15}$

C. $\frac{13}{15}$

B. $\frac{13}{30}$

D. $\frac{14}{75}$

D

Use for questions 11-12: A basket contains 8 green apples, 11 red apples, 5 nectarines, and 12 oranges. Mark randomly chooses a piece of fruit, eats it, then selects another. Find each probability.

11. $P(\text{a green apple then a nectarine})$

$$\frac{8}{36} \cdot \frac{5}{35}$$

A. $\frac{23}{63}$

C. $\frac{13}{36}$

B. $\frac{5}{162}$

D. $\frac{2}{63}$

D

12. $P(\text{two oranges})$

$$\frac{12}{36} \cdot \frac{11}{35}$$

A. $\frac{1}{9}$

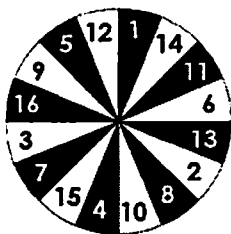
C. $\frac{11}{105}$

B. $\frac{4}{35}$

D. $\frac{68}{105}$

C

13. The spinner below is spun. Find the probability that it lands on an odd number, given that it lands on a white space. Write your answer as a fraction in simplest form.



$$\frac{3}{8}$$

$\frac{3}{8}$

14. Two dice are rolled. What is the probability that the sum of the dice is at least 5, given that one die is showing a 2.

$$\frac{4}{6}$$

A. $\frac{1}{2}$

C. $\frac{1}{6}$

B. $\frac{2}{3}$

D. $\frac{1}{3}$

B

Use for questions 15-16. Out of a group of 120 students that were surveyed about winter sports, 28 said they ski and 52 said they snowboard. Sixteen of the students who said they ski said they also snowboard. If a student is chosen at random, find each probability.

15. $P(\text{snowboard} | \text{ski})$

$$\frac{16}{28}$$

A. $\frac{4}{23}$

C. $\frac{4}{13}$

B. $\frac{4}{17}$

D. $\frac{4}{7}$

D

16. $P(\text{ski} | \text{does not snowboard})$

$$\frac{12}{68}$$

A. $\frac{3}{17}$

C. $\frac{3}{7}$

B. $\frac{7}{17}$

D. $\frac{3}{13}$

A

Use for questions 17-18. The table below shows the number of students at Parkland High School that made honor roll last quarter. If a student is chosen at random, find each probability. Give your answers as fractions in simplest form.

	Honor Roll	
	Yes	No
9 th Grade	72	120
10 th Grade	95	117
11 th Grade	76	124
12 th Grade	57	139

17. $P(\text{honor roll} \mid 11^{\text{th}} \text{ grader})$

$$\frac{76}{200}$$

$$\frac{19}{50}$$

18. $P(9^{\text{th}} \text{ grader} \mid \text{not honor roll})$

$$\frac{120}{500}$$

$$\frac{6}{25}$$

Use for questions 19-20. Rob and Casey are at batting practice. The table below shows the joint relative frequencies for their misses, fouls, hits, and homeruns. Give the marginal relative frequencies, then find each probability.

	Rob	Casey	Total
Misses	.05	.1125	.1625
Fouls	.10	.05	.15
Hits	.2875	.3375	.625
Homeruns	.0375	.025	.0625
Total	.475	.525	1

19. $P(\text{homerun} \mid \text{Casey})$

$$\frac{.025}{.525}$$

A. $\frac{1}{40}$ B. $\frac{2}{5}$ C. $\frac{1}{21}$ D. $\frac{2}{15}$

C

20. $P(\text{Rob} \mid \text{hit})$

$$\frac{.2875}{.625}$$

A. $\frac{15}{32}$ B. $\frac{23}{80}$ C. $\frac{23}{50}$ D. $\frac{23}{38}$

C

21. The odds of winning a game at the carnival are 3 in 7. If Liza plays the game nine times, what is the probability that she wins exactly two times?

$${}^9C_2 \left(\frac{3}{7}\right)^2 \left(\frac{4}{7}\right)^7$$

- A. 13.15%
B. 3.12%
C. 10.84%
D. 8.23%

A

22. When Sophia's mom picks her up from school, she is late one third of the time. For the next 15 days, what is the probability that she will be late at most three days?

$${}^{15}C_3 \left(\frac{1}{3}\right)^3 \left(\frac{2}{3}\right)^{12}$$

- A. 79.08%
B. 79.31%
C. 20.69%
D. 20.92%

D

23. When Tom plays darts, he hits the target 65% of the time. Find the probability that he hits the target at least four out of next six attempts.

$${}^6C_4 (.65)^4 (.35)^2$$

- A. 57.17%
B. 64.71%
C. 42.83%
D. 35.29%

B

24. What is the fourth term in the binomial expansion of $(2x - y)^7$?

$${}^7C_3 (2x)^4 (-y)^3$$

- A. $280x^3y^4$
B. $560x^4y^3$
C. $-280x^3y^4$
D. $-560x^4y^3$

D

Use for questions 25-26: The following data represents the high temperatures for the next several days in Buffalo, NY: {17, 26, 3, 13, 29, 34, 32, 30, 41}

- 25.** Find the mean absolute deviation. Round your answer to the nearest hundredth of a degree.

$$\frac{8 + 1 + 22 + 12 + 4 + 9 + 7 + 5 + 16}{9}$$

9.33°

- 26.** Which of the following values represent the standard deviation and variance for the data?

- A. Standard Deviation = 9.33; Variance = 3.05
- B. Standard Deviation = 9.33; Variance = 87.05
- C. Standard Deviation = 11.16; Variance = 124.55
- D. Standard Deviation = 11.16; Variance = 3.34

C

- 27.** Given Data Set A: {12, 14, 17, 5, 10, 8}
If 6 is added to this data set, which statement accurately describes the change in the disbursement of data.

$$A = \sigma = 3.9$$

$$\sigma = 4.03$$

- A. It will decrease.
- B. It will increase.
- C. It will stay the same.
- D. It's impossible to tell.

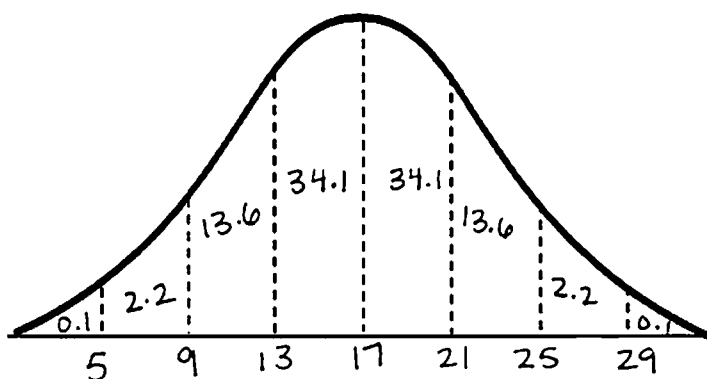
B

- 28.** If the mean and standard deviation of a data set are given, which data set has a variance equal to 16?

- A. Mean = 4; Standard Deviation = 8
- B. Mean = 256; Standard Deviation = 8
- C. Mean = 4; Standard Deviation = 256
- D. Mean = 256; Standard Deviation = 4

D

The wind speeds throughout the month of April were normally distributed with a mean of 17 mph and a standard deviation of 4 mph. Label the normal distribution curve, then answer questions 29-32:



- 29.** What percent of the days had wind speeds between 5 and 13 mph?

15.8%

- 30.** What percent of the days had wind speeds of at least 9 mph?

97.7%

- 31.** What is the probability that a day had a wind speed of no more than 21 mph?

84.1%

- 32.** Approximately how many days had wind speeds between 17 and 25 mph?

14 days

- 33.** The time it takes to complete a test is normally distributed with a mean of 45 minutes and a standard deviation of 12 minutes. If Nick took an hour and 8 minutes to complete the test, find his z -score to the nearest hundredth of a second.

$$z = \frac{68 - 45}{12}$$

1.92

- 34.** The mean price of a computers sold at a particular store is \$949 with a standard deviation of \$75. The z -score of a certain computer is -0.8. What is the price of the computer?

$$-0.8 = \frac{x - 949}{75}$$

\$889

- 35.** The number of ounces of water a person drinks per day is normally distributed with a standard deviation of 15 ounces. If Sean drinks 88 ounces per day with a z -score of 1.6, what is the mean ounces a day of water that a person drinks?

$$1.6 = \frac{88 - \mu}{15}$$

64 oz

- 36.** The scores on an Algebra 2 midterm were normally distributed with a mean of 82. If Kendall scored 67 with a z -score of -2.5, what is the standard deviation?

$$-2.5 = \frac{67 - 82}{\sigma}$$

6

Use for questions 37-40: The 100-meter dash times in the girls track meet were normally distributed with a mean of 13 seconds and a standard deviation of 0.3 seconds.

- 37.** What is the probability that a runner finished between 12.4 and 14 seconds?

$$\frac{12.4 - 13}{0.3} = -2$$

$$\frac{14 - 13}{0.3} = 3.33$$

97.68%

- 38.** What is the probability that a runner finished in under 12.7 seconds?

$$\frac{12.7 - 13}{0.3} = -1$$

15.87%

- 39.** What is the probability that it took a runner more than 13.5 seconds?

$$\frac{13.5 - 13}{.3} = 1.67$$

4.75%

- 40.** Lana finished the race in 13.2 seconds. If 84 other girls ran in the event, approximately how many runners did she beat?

$$\frac{13.2 - 13}{.3} = 0.67$$

25.14%

21 runners

CREDITS

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Many thanks to these
talented artists!